

Q1 Consider the grammar :

$S \rightarrow A$

$A \rightarrow \$B\$ \mid id$

$B \rightarrow B, A \mid A$

If $I_0 = \text{CLOSURE}(\{[S \rightarrow A]\})$ then, how many items be in the set for $\text{GOTO}(I_0, \$)$

- (A) 3 (B) 4
(C) 5 (D) 6

Q2 In a schema $R(A, B, C, D, E, F, G, H)$, each field of R contains only atomic values.

$F = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$ is a set of functional dependencies F^+ is exactly the set of FDs that holds R . The relation R is :

- (A) In 1NF, but not in 2 NF
(B) In 2 NF, but not in 3 NF
(C) In 3NF, but not in BCNF
(D) In BCNF

Q3 Which of the following statements about pointers in C are TRUE.

- (A) Pointers can be used to access array elements
(B) Pointers can store the address of another pointer
(C) Pointers are automatically dereferenced in expression
(D) Pointers cannot be used to access structure members

Choose the correct answer from the options given below :

- (A) (A) and (C) Only
(B) (A) and (B) Only
(C) (B) and (C) Only
(D) (C) and (D) Only

Q4 In a token ring network the transmission speed is 10^7 and the propagation speed is $200 \text{ m}/\mu\text{s}$. The 1-bit delay in this network is equivalent to

- (A) 20 m (B) 30 m
(C) 50 m (D) 40 m

Q5 Which of the following statements are TRUE about mutual exclusion in concurrent

programming ?

(A) Mutual exclusion ensures that only one process can be in a critical section at any given time.

(B) Mutual exclusion are designed to prevent conflicts and ensure that only one process can access shared resources at a time.

(C) Mutual exclusion can use various algorithms to ensure that processes do not enter the critical section simultaneously.

(D) Mutual exclusion allows multiple processes to access the critical section simultaneously to improve performance.

Choose the correct answer from the options given below :

- (A) (A), (B), (C) Only
(B) (B), (C), (D) Only
(C) (B), (D), (A) Only
(D) (A), (C), (D) Only

Q6 Consider the function in C code :

```
Cal(a, b) {
    if( b!=1 ) {
        if( a!=1 ) {
            printf("*");
            Cal(a/2,b);
        }
    }
    else {
        b=b-1;
        Cal(10,b);
    }
}
```

How many times * is going to be printed, if the function is called with $\text{Cal}(10, 10)$; ?

- (A) 25 (B) 23
(C) 24 (D) 27

Q7 Consider the following relations on $\{1,2,3,4\}$.

Which of the following relations are reflexive ?

- (A) $R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$
(B) $R_2 = \{(1,1), (1,2), (2,1)\}$
(C) $R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$



(D) $R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$

(E) $R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$

Choose the correct answer from the options given below :

(A) (A), (B) and (C) Only

(B) (A), (D) and (E) Only

(C) (D) and (E) Only

(D) (C) and (E) Only

Q8 Consider a relation schema $R = (U, V, W, X, Y, Z)$, on which the following functional dependencies hold

$\{U \rightarrow V, VW \rightarrow X, Y \rightarrow W; X \rightarrow U\}$

The candidate keys of R are :

(A) UY, VY

(B) UY, VY, XY

(C) UYZ, VYZ, VWZ

(D) UYZ, VYZ, XYZ

Q9 Which of the following is correct way to declare a functional pointer in C ?

(A) `int *func (int, int);`

(B) `int (*func) (int, int);`

(C) `int (func*) (int, int);`

(D) `int *func* (int, int);`

Q10 Arrange the following steps of Feature Drive Development (FDD) process in the correct sequence :

(A) Develop an overall model

(B) Build by feature

(C) Plan by feature

(D) Design by feature

(E) Build a feature list

Choose the correct answer from the options given below :

(A) (A), (C), (B), (E), (D)

(B) (A), (E), (C), (D), (B)

(C) (B), (A), (D), (E), (C)

(D) (A), (B), (C), (E), (D)

Q11 Consider the given number $(45)_y$ where y is the base of the number. Some of the possible values of y are given below.

(A) 5

(B) 6

(C) 7

(D) 8

Choose the correct answer from the options given below :

(A) (A), (B) and (C) Only

(B) (B), (C) and (D) Only

(C) (A), (C) and (D) Only

(D) (A), (B), (C) and (D)

Q12 Consider the following C program :

```
#include<stdio.h>
int main() {
    int x[] = {2,4,6,8,10};
    int a, b=0, *y=x+4;
    for( a=0; a<5; a++) {
        b = b +(*y-a) - *(y-a);
    }
    printf("%d\n",b);
    return 0;
}
```

What will be the output of the above C program ?

(A) 4

(B) 6

(C) 8

(D) 10

Q13 Match List - I with List - II.

List - I	List - II
(A) The activation function	(I) is called the delta rule.
(B) The learning method of perceptron	(II) is one of the key components of the perceptron as in the most common neural network architecture.
(C) Areas of application of artificial neural	(III) is always boolean like a switch. network include
(D) The output of the perceptron	(IV) system identification and control.

Choose the correct answer from the options given below :

(A) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

(B) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(C) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)



(D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q14 Match List - I with List - II.

List - I	List - II
(A) Representation of bits	(I) Transport layer
(B) Physical Address	(II) Physical layer
(C) Logical Address	(III) Network layer
(D) Segmentation and reassembly	(IV) Data link layer

Choose the correct answer from the options given below :

(A) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(B) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

(C) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(D) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

Q15 Which of the following are Agile Process Models ?

(A) Extreme Programming (XP)

(B) Waterfall

(C) Scrum

(D) Spiral

(E) Incremental

Choose the correct answer from the options given below :

(A) (A) and (C) Only

(B) (B) and (C) Only

(C) (A) and (D) Only

(D) (B) and (E) Only

Q16 Which of the following statements are TRUE about Privileged Instructions ?

(A) It can only be executed by the Operating System kernel and not by user applications.

(B) It is designed to perform operations that can directly affect the hardware or system state such as I/O operations or changing memory management setting.

(C) User applications can execute privileged instructions if they have to correct permissions, set by the Operating System.

(D) It usually executed in user mode to ensure the safety and security of the system.

Choose the correct answer from the options given below :

(A) (A) and (B) Only

(B) (A), (B) and (C) Only

(C) (B) and (C) Only

(D) (B), (C) and (D) Only

Q17 Consider a CSMA/CD network that transmits data at the rate of 100 Mbps over a 1 Kilometre cable with no repeater. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable.

(A) 8000

(B) 16000

(C) 12000

(D) 20000

Q18 Consider the Grammar:

$T \rightarrow Qx$

$Q \rightarrow RS$

$R \rightarrow y \mid \epsilon$

$S \rightarrow z \mid \epsilon$

Here x, y, z are terminals and T, Q, R, S are non-terminals.

What will be the follow set of the non-terminal R?

(A) {x,y}

(B) {y,z}

(C) {z,x}

(D) {ε}

Q19 Host A (on TCP/IPv4 Networks) send an IP Datagram D to host B (also on TCP/IPv4 network). Assume that no error occurred during transmission of D, when D reaches B. Which of following header field may differ from that of original datagram D ?

(A) TTL

(B) Checksum

(C) Fragment offset

(D) Source IP

(E) Destination IP

Choose the correct answer from the options given below :

(A) (A) and (B) Only

(B) (A), (B) and (C) Only

(C) (A), (B), (C) and (D) Only

(D) (A), (B), (C) and (E) Only

Q20 Which of the following properties correctly describe a Regular Grammar ?



(A) All production rules are of the form $A \rightarrow xB$ or $A \rightarrow x$, where A and B are non terminal symbols and x is a terminal symbol.

(B) Regular grammars are more powerful than context-free grammars and can express any type of language.

(C) There is a direct correspondence between regular grammar and finite automata.

(D) Regular grammars can generate languages that are not recognised by any type of automata.

Choose the correct answer from the options given below :

(A) (A) and (B) Only

(B) (B) and (C) Only

(C) (C) and (D) Only

(D) (A) and (C) Only

Q21 Consider a schema $R(P, Q, R, S)$ and the following functional dependencies $P \rightarrow Q, Q \rightarrow R, R \rightarrow S, S \rightarrow Q$. Then decomposition of $R_1(P, Q), R_2(Q, R)$ and $R_3(Q, S)$ is :

(A) Dependency Preserving and lossless join

(B) Lossless Join but not dependency preserving

(C) Dependency preserving but not lossless Join

(D) Not dependency preserving and not lossless join

Q22 Arrange the following phases of the Agile process in the correct sequence :

(A) Design

(B) Release

(C) Testing

(D) Development

(E) Planning

Choose the correct answer from the options given below :

(A) (A), (B), (D), (E), (C)

(B) (A), (D), (E), (B), (C)

(C) (E), (A), (D), (C), (B)

(D) (E), (D), (A), (C), (B)

Q23 Which of the following are TRUE about constructors in C++ ?

(A) A constructor can be overloaded.

(B) A constructor does not have a return type.

(C) A constructor must be declared as a friend

function.

(D) A constructor is called when an object is destroyed.

Choose the correct answer from the options given below :

(A) (B), (C), (D) Only

(B) (B), (C) Only

(C) (A), (B), (C) Only

(D) (A), (B) Only

Q24 Let P be "It is hot day" and q be "The temperature is 48°C ". Write in simple sentences the meaning of $\neg p \wedge \neg q$.

(A) It is hot day or temperature is 48°C

(B) It is cold day or temperature is 48°C

(C) It is neither a hot day nor temperature is 48°C

(D) It is not a hot day

Q25 If a software project has 2000 lines of code (LOC) and the average productivity rate is 10 LOC per person-hour, how many person-hours are required to complete the project?

(A) 100

(B) 150

(C) 200

(D) 250

Q26 Arrange the following steps in proper sequence involved in a Genetic Algorithm :

(A) Selection

(B) Initialization

(C) Crossover

(D) Mutation

(E) Evaluation

Choose the correct answer from the options given below :

(A) (A), (B), (C), (D), (E)

(B) (E), (A), (B), (D), (C)

(C) (B), (E), (A), (C), (D)

(D) (A), (C), (B), (D), (E)

Q27

```
#include<stdio.h>
void f1(char *x, char *y) {
    char *t1;
    t1 = x;
    x = y;
    y = t1;
}
```



```

void f2(char **x, char **y) {
    char *t1;
    t1 = *x;
    *x = *y;
    *y = t1;
}

int main() {
    char *a = "ONE", *b = "TWO";
    f1(a, b);
    printf("%s %s ", a, b);
    f2(&a, &b);
    printf("%s %s", a, b);
    return 0;
}

```

- (A) ONE TWO TWO ONE
 (B) TWO ONE ONE TWO
 (C) ONE TWO ONE TWO
 (D) TWO ONE TWO ONE

Q28 What is SQL primarily used for in the context of relational databases ?

- (A) To design user interfaces
 (B) To create and manipulate databases
 (C) To display data on web pages
 (D) To format printed reports

Q29 Which of the following C++ statements correctly declares an abstract class ?

- (A) class A { virtual void show ()=0 ; };
 (B) class A { void show ()=0 ; };
 (C) class A { void show() { } ; };
 (D) class A { show ()=0; };

Q30 Arrange the following recurrence relations in increasing order of their time capacity:

- (A) $T(n) = T(n/2) + 1$
 (B) $T(n) = 2T(n/2) + n$
 (C) $T(n) = 3T(n/3) + n$
 (D) $T(n) = 2T(n/2) + \sqrt{n}$
 (E) $T(n) = T(n-1) + 1$

Choose the correct answer from the options given below:

- (A) (E), (A), (B), (D), (C)
 (B) (A), (E), (D), (B), (C)
 (C) (E), (A), (D), (B), (C)

(D) (A), (B), (D), (E), (C)

Q31 How does a relational database ensure data integrity ?

- (A) By encrypting all data stored
 (B) By enforcing rules defined in the schema
 (C) By compressing data for efficient storage
 (D) By allowing unrestricted access to all users

Q32 Match List - I with List - II.

List - I (IP Address)	List - II (Class)
(A) 10.20 .30 .40	(I) Class E
(B) 210.20 .30 .3	(II) Class B
(C) 180.30 .100 .10	(III) Class A
(D) 252.5 .15 .11	(IV) Class C

Choose the correct answer from the options given below :

- (A) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
 (B) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)
 (C) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
 (D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q33 Find the correct sequence of the storage devices in descending order based on their access time.

- (A) Registers
 (B) Magnetic Disk
 (C) Magnetic Tapes
 (D) Main memory
 (E) Optical Disk

Choose the correct answer from the options given below :

- (A) (B), (A), (D), (C), (E)
 (B) (B), (A), (E), (D), (C)
 (C) (C), (B), (A), (E), (D)
 (D) (C), (E), (B), (D), (A)

Q34 A graph G with number of vertices greater and equal than three i.e. ($n \geq 3$) is a Hamiltonian graph, if the degree of each vertex is greater and equal to

- (A) Equal to number of vertices
 (B) Double of number of vertices
 (C) Half of number of vertices
 (D) Four times of number of vertices



Q35 Fifth normal form is concerned with :

- (A) Join Dependency
- (B) Domain-Key
- (C) Multivalued dependency
- (D) Functional dependency

Q36 Consider the transactions T_1, T_2, T_3 , and the schedules S_1 and S_2 given below.

$T_1: r_1(x); r_1(z); w_1(z)$

$T_2: r_2(y); r_2(z); w_2(z)$

$T_3: r_3(y); r_3(x); w_3(y)$

$S_1: r_1(x); r_3(y); r_3(x); r_2(y); r_2(z); w_3(y); w_3(z);$
 $r_1(z); w_1(x); w_1(z)$

$S_2: r_1(x); r_3(y); r_2(y); r_3(x); r_1(z); r_2(z); w_3(y);$
 $w_1(x); w_2(z); w_1(z)$

Which one of the following statements about the schedules is TRUE?

(A) Only S_1 is conflict-serializable

(B) Only S_2 is conflict-serializable

(C) Both S_1 and S_2 are conflict-serializable

(D) Neither S_1 nor S_2 is conflict-serializable

Q37 A CPU has a 5-stage pipeline with the following stages Fetch (F), Decode (D), Execute (E), Memory (M) and Write-back (W). Each stage takes one clock cycle to complete. Assume there are no pipeline stalls and the pipeline is initially empty. How many clock cycles are required to complete the execution of 10 instructions?

- (A) 10
- (B) 14
- (C) 15
- (D) 19

Q38 Match List - I with List - II.

List - I (computing systems)	List - II (working/output)
(A) Half Adder	(I) Has n input and 2^n output
(B) Decoder	(II) CPU Storage Unit
(C) Register	(III) Used to store program at runtime
(D) Main memory	(IV) 2-bit addition circuit

Choose the correct answer from the options given below:

- (A) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
- (B) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
- (C) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
- (D) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

Q39 Arrange the following steps in a proper sequence for the process of training a neural network.

- (A) Weight initialization
- (B) Feed forward
- (C) Back Propagation
- (D) Loss Calculation
- (E) Weight Update

Choose the correct answer from the options given below :

- (A) (A), (B), (D), (C), (E)
- (B) (D), (B), (A), (C), (E)
- (C) (A), (C), (D), (B), (E)
- (D) (E), (C), (B), (D), (A)

Q40 Consider the following statements regarding combinational and sequential circuits.

- (A) Output of combinational circuits depends on the only current input.
- (B) Output of combinational circuit depends on the both current input and previous output.
- (C) Output of sequential circuit depends on the current input.
- (D) Output of sequential circuit depends on both current input and previous output.

Choose the correct answer from the options given below :

- (A) (A) and (C) Only
- (B) (A) and (D) Only
- (C) (B) and (D) Only
- (D) (B) and (C) Only

Q41 Which of the following relations can not be decomposed in to BCNF with a lossless join and dependency-preserving decomposition?

- (A) $R(V, W, X, Y, Z) \{VW \rightarrow X, WX \rightarrow V, VX \rightarrow W, W \rightarrow Y, Y \rightarrow Z\}$
- (B) $R(V, W, X, Y, Z) \{VW \rightarrow W, X \rightarrow Y\}$
- (C) $R(V, W, X, Y, Z) \{VW \rightarrow X, X \rightarrow VY\}$
- (D) $R(V, W, X, Y, Z) \{VW \rightarrow X, X \rightarrow Y, Y \rightarrow Z, Z \rightarrow V\}$



Q42 `#include<iostream.h>`
`using namespace std;`
`void swap(int &x, int &y)`
`{`
`int temp=x;`
`x=y;`
`y=temp;`
`}`
`int main()`
`{`
`int a=5, b=10;`
`swap(a,b);`
`swap(a,b);`
`cout<<a<<" "<<b;`
`return 0;`
`}`

What will be the output of the above code?

- (A) 5, 10 (B) 10, 5
 (C) 5, 5 (D) 10, 10

Q43 Which of the following are typical activities in the software process lifecycle?

- (A) Requirement Analysis
 (B) System Design
 (C) Code Refactoring
 (D) Deployment
 (E) Substructure

Choose the correct answer from the options given below :

- (A) (A), (B), (C) Only
 (B) (B), (C), (D) Only
 (C) (A), (B), (D) Only
 (D) (A), (D), (E) Only

Q44 Which of the following best describes the structure of a relational database ?

- (A) Data organized into tables with rows and columns
 (B) Data organized into files and folders
 (C) Data organized into a hierarchical tree structure
 (D) Data organized into a network of interconnected nodes

Q45 A coin is tossed successively three times. Find the Probability (P), Event (E), Sample space (S) of

getting exactly one head or two heads, where n is number of occurrence.

- (A) $n(S) = 8$ and $n(E) = 4$
 (B) $n(E) = 6$ and $n(S) = 8$
 (C) $P(E) = 3/4$
 (D) $P(E) = 1/2$

Choose the correct answer from the options given below :

- (A) (A), (B), (C) and (D)
 (B) (B) and (C) Only
 (C) (A) and (D) Only
 (D) (C) and (D) Only

Q46 Out of the following steps in the proper sequence for simplifying a Boolean function using a Karnaugh map (K-map).

- (A) Identify and group the largest possible cluster of 1's
 (B) Draw the K-map for the given Boolean function
 (C) Write the simplified Boolean expression from the grouped clusters
 (D) Transfer the truth table values to the K-map

Choose the correct answer from the options given below :

- (A) (B), (D), (A), (C)
 (B) (D), (B), (A), (C)
 (C) (B), (A), (D), (C)
 (D) (A), (B), (C), (D)

Q47 Consider the following if p and q are two statements.

- (A) $\sim(p \wedge q) \equiv \sim p \vee \sim q$
 (B) $\sim(p \vee q) \equiv \sim p \wedge \sim q$
 (C) $p \wedge \sim p \equiv T$
 (D) $\sim(p \Rightarrow q) \equiv p \wedge \sim q$
 (E) $p \vee q \equiv \sim p \vee \sim q$

Choose the correct answer from the options given below :

- (A) (A), (B) and (D) only
 (B) (A), (C) and (D) only
 (C) (C), (D) and (E) only
 (D) (A), (B) and (C) only

Q48 What is the output of the code given below :

`#include<iostream.h>`



```
using namespace std;
class Base {
public:
Base(){
    cout<<"Base Const";
}
virtual ~Base() {
    cout<<"Base dest";
}
};
class Derived : public Base {
public:
Derived() {
    cout<<"Derived Const";
}
~Derived() {
    cout<<"Derived dest";
}
};
int main() {
Base *b = new Derived();
delete b;
return 0;
}
```

- (A) Base Const Derived Const Derived dest Base dest
 (B) Base Const Derived Const Base dest Derived dest
 (C) Derived Const Base Const Base dest Derived dest
 (D) Base Const Derived Const Base dest

Q49 If L_1 and L_2 are context free languages, which one of the following is true about $L_1 \cap L_2$?

- (A) $L_1 \cap L_2$ is context free.
 (B) $L_1 \cap L_2$ is Regular
 (C) $L_1 \cap L_2$ is Recursive Enumerable
 (D) $L_1 \cap L_2$ is Context Sensitive

Q50 Consider the following C code:

```
#include<stdio.h>
int temp =0;
int fun(int x, int y)
{
    int z; temp++;
```

```
if(y==3) return (x*x*x);
else {
    z=fun(x,y/3);
    return (z*z*z);
}
}
```

```
int main() {
    fun(4,81);
    printf("%d",temp);
}
```

what will be the output of the following code :

- (A) 3 (B) 4
 (C) 5 (D) 6

Q51 Which of the following tasks could be attained using syntax trees in compiler design ?

- (A) Type Checking
 (B) Code Generation
 (C) Code Optimization
 (D) Error Handling

Choose the correct answer from the options given below :

- (A) (A), (B), (C) Only
 (B) (B), (C), (D) Only
 (C) (A), (C), (D) Only
 (D) (A), (B), (D) Only

Q52 Match List - I with List - II.

List - I	List - II
(A) HTML	(I) Allow for dynamic and interactive web pages.
(B) DHTML	(II) Defines the structure of web pages.
(C) XML	(III) Object-oriented programming language for web applications.
(D) JAVA	(IV) Used for data storage and transport.

Choose the correct answer from the options given below :

- (A) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
 (B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
 (C) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
 (D) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)



Q53 Match List - I with List - II.

List - I (Counter)	List - II (uses/working)
(A) N-bit Ring Counter	(I) Uses universal clock
(B) Synchronous Counter	(II) Counts exactly N states
(C) Asynchronous Counter	(III) Counts 0 to 9
(D) Decimal Counter	(IV) Main clock is applied to first flip-flop only

Choose the correct answer from the options given below:

- (A) (A)-(II), (B)-(IV), (C)-(I), (D)-(III)
 (B) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
 (C) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)
 (D) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

Q54 Arrange the following steps of file handling in C in the correct order :

- (A) Close the file
 (B) Read from or write to the file
 (C) Open the file
 (D) Check for error

Choose the correct answer from the options given below :

- (A) (A), (C), (D), (B)
 (B) (D), (B), (C), (A)
 (C) (C), (D), (B), (A)
 (D) (B), (D), (A), (C)

Q55 Match List - I with List - II.

List - I (Algorithms)	List - II (Characteristics)
(A) First Come First Served (FCFS)	(I) Each process is assigned a priority.
(B) Shortest Job First (SJF)	(II) Ensures fair allocation of CPU time by assigning time slice.
(C) Round Robin (RR)	(III) Processes are executed in the order they arrive.
(D) Priority Scheduling	(IV) Select the process for execution with

	smallest next Burst time.
--	---------------------------

Choose the correct answer from the options given below :

- (A) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
 (B) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)
 (C) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
 (D) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

Q56 Arrange the following steps in a proper sequence for the typical process of a DNS query :

- (A) Query authoritative DNS Server
 (B) Query local DNS Server
 (C) Local DNS Server Check Cache
 (D) Query root DNS Server
 (E) Query TLD DNS Server

Choose the correct answer from the options given below :

- (A) (A), (C), (E), (B), (D)
 (B) (E), (A), (C), (B), (D)
 (C) (A), (B), (C), (D), (E)
 (D) (B), (C), (D), (E), (A)

Q57 Match List - I with List - II.

List - I (operating system concepts)	List - II (characteristics)
(A) Paging	(I) Evicts least recently used process
(B) LRU (Least Recently Used)	(II) Extends physical memory
(C) C-SCAN	(III) Logical to physical mapping
(D) Virtual Memory	(IV) Circular disk access

Choose the correct answer from the options given below :

- (A) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
 (B) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
 (C) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)
 (D) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

Q58 A class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

- (A) 1022 (B) 1023



(C) 2046

(D) 2047

Q59 Select the Sorting Algorithms that are stable.

- (A) Quick Sort
- (B) Bubble Sort
- (C) Insertion Sort
- (D) Merge Sort
- (E) Shell Sort

Choose the correct answer from the options given below :

- (A) (A), (B), (C) and (E) Only
- (B) (A), (D) and (E) Only
- (C) (B) and (C) Only
- (D) (B), (C) and (D) Only

Q60 _____ is a Self Balancing binary search tree, where the path from the root to the furthest leaf is no more than twice as long as the path from the root to nearest leaf.

- (A) Expression tree
- (B) Game tree
- (C) Red-Black tree
- (D) Threaded tree

Q61 Let $L(x, y)$ be the statement "x loves y" where the domain for both x and y consists of all people in the world. Use quantifiers to express "Joy is loved by everyone".

- (A) $\forall x L(x, \text{Joy})$
- (B) $\forall y L(\text{Joy}, y)$
- (C) $\exists y \forall x L(x, y)$
- (D) $\exists x \neg L(\text{Joy}, x)$

Q62 What will be the output of the following C code?

```
#include<stdio.h>
void main()
{
    int arr[5] = {10,20,30,40,50};
    int *p = (int*) (&arr +1);
    printf("%d%d", *(arr+1), *(p-1));
}
```

- (A) 10 50
- (B) 20 50
- (C) 30 40
- (D) 20 40

Q63 Arrange the following stages of a Turing Machine (TM) operation in the correct order as they occur during computation.

- (A) Writing a symbol on the tape
- (B) Moving the tape head left to right
- (C) Reading a symbol from the tape
- (D) Transitioning to a new state based on the current state and symbol read
- (E) Halting and accepting or rejecting the input

Choose the correct answer from the options given below :

- (A) (C), (A), (B), (D), (E)
- (B) (C), (B), (A), (D), (E)
- (C) (C), (D), (A), (B), (E)
- (D) (C), (D), (B), (A), (E)

Q64 Match List - I with List - II.

List - I (software design principles)	List - II (Definition)
(A) Cohesion	(I) Degree to which one module relies on another module.
(B) Coupling	(II) Dividing a software system into distinct modules.
(C) Abstraction	(III) Degree to which elements of a module belong together.
(D) Modularity	(IV) Simplifying complex reality by modeling classes appropriate to the problem.

Choose the correct answer from the options given below :

- (A) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
- (B) (A)-(II), (B)-(III), (C)-(IV), (D)-(I)
- (C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
- (D) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Q65 Which of the following languages can be recognized by Non-Deterministic Finite Automata(NFA) but cannot be recognized by Deterministic Finite Automata(DFA)?

- (A) $L_1 = \{w \in \{0,1\}^* \mid \text{the length of } w \text{ is even}\}$
- (B) $L_2 = \{w \in \{0,1\}^* \mid \text{the length of } w \text{ is odd}\}$
- (C) $L_3 = \{w \in \{0,1\}^* \mid \text{all 0's come before all 1's in } w\}$



(D) $L_4 = \{w \in \{0,1\}^* \mid w \text{ contains an equal number of 0's and 1's}\}$

(E) $L_5 = \{w \in \{0,1\}^* \mid \text{all 1's come before all 0's in } w\}$

(A) (A) and (B) Only

(B) (B) and (C) Only

(C) (C) and (D) Only

(D) None of the above

Q66 Given a project with an estimated effort of 1500 person-hours and a team of 5 people, how many days will it take to complete the project, if each person works 8 hours a day?

(A) 30

(B) 37.5

(C) 40

(D) 45.5

Q67 Arrange the following phases of database design in the correct order :

(A) Physical Design

(B) Conceptual Design

(C) Logical Design

(D) Requirement Analysis

Choose the correct answer from the options given below :

(A) (B), (D), (A), (C)

(B) (C), (A), (B), (D)

(C) (D), (B), (C), (A)

(D) (A), (D), (C), (B)

Q68 Consider the three way hand shaking process followed during TCP connection establishment between two hosts A and B . Let S and R be two random 32 -bit starting sequence numbers chosen by A and B, respectively. Suppose A sends a TCP segment having SYN bit =1, SEQ number = S and ACK bit =0 and B accepts the connection request.

Which one of the following choices represents the information present in the TCP segment header that is sent by B to A ?

(A) SYN bit =1, SEQ number =S+1, ACK bit =0, ACK number =R, FiN bit =0

(B) SYN bit =0, SEQ number =R, ACK bit =1, ACK number =S+1, FiN bit =0

(C) SYN bit =1, SEQ number =R, ACK bit =1, ACK number =S+1, FiN bit =0

(D)

SYN bit =1, SEQ number = R, ACK bit =1, ACK number =S, FiN bit =0

Q69 Two computers A and B are configured as follows : A has IP address 203.197.2.53 and subnet mask 255.255.128.0, B has IP address 203.197.75.201 and subnet mask 255.255.192.0. What one of the following statement is true?

(A) A and B both assume they are on the same network

(B) B assumes, A is on same network, A assumes B is on different network

(C) A assumes, B is on same network, but B assumes, A is on a different network

(D) A and B both assume they are on different network

Q70 In Software configuration management, what is the primary purpose of version control ?

(A) To control the changes made to software

(B) To document user requirements

(C) To estimate project cost

(D) To design the software architecture

Q71 Arrange the following stages of parsing in the correct order as they typically occur in the compilation process.

(A) Lexical Analysis

(B) Sematic Analysis

(C) Syntax Analysis

(D) Intermediate Code Generation

(E) Code Optimization

Choose the correct answer from the options given below :

(A) (A), (B), (C), (D), (E)

(B) (A), (C), (B), (D), (E)

(C) (A), (D), (B), (C), (E)

(D) (A), (C), (D), (B), (E)

Q72 Match List - I with List - II.

List - I (Queries)	List - II (Probability)
(A) A bag contains 6 white and 4 red balls. Two balls are drawn at random. What is the	(I) $3/68$



chance they will the same color?	
(B) In a pack of 52 cards, one card is drawn at random, What is the probability that it is either a king or queen?	(II) 14/68
(C) A bag contains 6 red, 4 white and 8 blue balls. If three balls are drawn at random, find the probability of 1 red and 2 white balls ?	(III) 2/13
(D) A bag contains 6 red, 4 white and 8 blue balls. If three balls are drawn at random. Find the probability of 2 blue and 1 red balls ?	(IV) 7/15

Choose the correct answer from the options given below :

- (A) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
 (B) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)
 (C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
 (D) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Q73 Match List - I with List - II.

List - I	List - II
(A) Natural language processing	(I) A method of training algorithm by rewarding desired behaviour and/or punishing undesired one.
(B) Reinforcement learning	(II) System designed to emulate the decision making abilities of a human expert.
(C) Support vector machine	(III) A branch of AI focused on understanding and generating human language.
(D) Expert system	(IV) A machine learning technique that finds the

	hyper plane that best separates different class in a feature space.
--	---

Choose the correct answer from the options given below :

- (A) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
 (B) (A)-(III), (B)-(II), (C)-(I), (D)-(IV)
 (C) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)
 (D) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

Q74 Match List - I with List - II.

List - I	List - II
(A) Dijkstra's Algorithms	(I) Find the shortest path between all pairs of vertices in a graph with positive or negative edge weights.
(B) Floyd-Warshall Algorithms	(II) Finds the shortest path in a weighted graph with non-negative edge weights.
(C) Bellman-Ford Algorithms	(III) Ford finds single-source shortest paths with possible negative weights
(D) Prim's Algorithms	(IV) Prim's algorithm finds the Minimum Spanning Tree (MST)

Choose the correct answer from the options given below :

- (A) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
 (B) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)
 (C) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)
 (D) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Q75 Which of the following statements are TRUE about Process Control Block (PCB) ?

- (A) The PCB contains information about the process state, such as whether it is running, waiting or terminated
 (B) The PCB includes the program code and data



segments of the process

(C) The PCB stores the process's memory management information, such as page tables and segment tables

(D) The PCB is used to track process scheduling information and CPU registers for process execution

Choose the correct answer from the options given below :

(A) (A), (B) and (C) Only

(B) (B), (C) and (D) Only

(C) (A), (C) and (D) Only

(D) (A) and (B) Only

Q76 #include<stdio.h>

void main()

{

int arr[] = {1, 2, 3, 4, 5};

int *p = arr;

printf("%d", *p++);

printf("%d", *(p+1));

}

Find the output of the above code?

(A) 1,2

(B) 1,3

(C) 2,3

(D) 1,4

Q77 Arrange the following Language Classes in ascending order according to their expressive power, as defined by Chomsky hierarchy:

(A) Context-free languages

(B) Context-sensitive languages

(C) Regular languages

(D) Unrestricted Grammars

Choose the correct answer from the options given below :

(A) (C), (A), (B), (D)

(B) (C), (A), (D), (B)

(C) (A), (C), (B), (D)

(D) (A), (D), (B), (C)

Q78 Let $G(V,E)$ be an undirected graph with l edges. Then the sum of degree of all vertices is equal to _____.

(A) $2l$

(B) $l/2$

(C) l^2

(D) $\sqrt{l}$

Q79 40 software professionals were interviewed for a job. 25 knew PYTHON 20 knew JAVA and 7 knew neither language. How many knew both languages ?

(A) 12

(B) 53

(C) 10

(D) 88

Q80 (A) In IPv4 addressing, a block of address can be define as $x.y.z.t/n$ in which $x.y.z.t$ define one of the address and $/n$ define the mask

(B) The first address in the block can be found by setting the rightmost $32-n$ bits to 0s

(C) Address of class C is used for multicast communication

(D) There are five classes in IPv4 address

(E) Supernetting combine several networks into one large network

Choose the correct answer from the options given below :

(A) (A), (B) and (C) Only

(B) (A), (B), (D) and (E) Only

(C) (A), (C), (D) and (E) Only

(D) (C), (D) and (E) Only

Q81 Match List - I with List - II.

List - I (Testing Type)	List - II (Description)
(A) Unit testing	(I) Testing individual components of the software.
(B) Integrating testing	(II) Testing the interaction between integrated components.
(C) System testing	(III) Testing to verify the system meets business needs.
(D) Acceptance testing	(IV) Testing the complete system to ensure it meets requirements.

Choose the correct answer from the options given below :

(A) (A)-(I), (B)-(II), (C)-(III), (D)-(IV)

(B) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(C) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)



(D) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

- Q82** Consider the following statement regarding special purpose registers.
- (A) Program Counter (PC) keeps the track of next instruction executed
- (B) Instruction register holds the address of first instruction to be executed
- (C) Accumulator holds the output of ALU
- (D) Program Counter (PC) keeps the track of only first instruction of the program
- Choose the correct answer from the options given below :
- (A) (A) and (C) Only
- (B) (A) and (D) Only
- (C) (B) and (D) Only
- (D) (A), (B) and (C) Only

- Q83** Arrange the following steps in the correct order to solve the Knapsack problem using Dynamic Programming.
- (A) Define the base case when the capacity is zero (0) or no items are left to consider
- (B) Compute the maximum value that can be obtained using items up to the i-th item and a knapsack capacity of 0
- (C) Identify subproblems and their dependencies based on items weights and values
- (D) Initialize a table to store results of subproblems
- (E) Iterate through each item and each possible Capacity to fill the table
- Choose the correct answer from the options given below :
- (A) (C), (D), (A), (E), (B)
- (B) (D), (C), (A), (E), (B)
- (C) (A), (C), (D), (E), (B)
- (D) (D), (A), (C), (E), (B)

- Q84** A multi processor system with 16 processors is used to execute a parallelizable task. If the serial portion of the task takes 200 clock cycles and the parallel portion take 800 clock cycles. When all 16 processor are used how many total clock cycles are required to complete the task ?
- (A) 250 (B) 300

(C) 400

(D) 450

- Q85** Arrange the given steps required for a Direct Memory Access (DMA) transfer in the correct order.
- (A) Initiate DMA transfer request
- (B) Transfer data directly between peripheral and memory
- (C) Processor grants DMA control over the system bus
- (D) DMA controller completes data transfer and signals completion
- Choose the correct answer from the options given below :
- (A) (C), (A), (B), (D)
- (B) (A), (C), (B), (D)
- (C) (A), (B), (C), (D)
- (D) (C), (B), (A), (D)

- Q86** Match List - I with List - II

List - I (Recurrence Relation)	List - II (Complexity)
(A) $T(n) = 2T(n/2) + n$	(I) $T(n) = \theta(n \log n)$ <i>{exact solution}</i>
(B) $T(n) = T(n/2) + 1$	(II) $O(n^2)$
(C) $T(n) = 2T(n/2) + 1$	(III) $T(n) = \theta(n)$ <i>{exact solution}</i>
(D) $T(n) = T(n-1) + n$	(IV) $O(n)$

- Choose the correct answer from the options given below:
- (A) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)
- (B) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)
- (C) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)
- (D) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

- Q87** Translate $\forall x \exists y (x < y)$ in English. Consider domain as a real number for both the variables.
- (A) For all real numbers x, there exists a real number y such that x is less than y
- (B) For every real numbers y, there exists a real number x such that x is less than y
- (C)



For some real numbers x , there exists a real number y such that x is less than y

(D) For each and every real numbers x and y , x is less than y

Q88 A vector processor with 16 lanes can perform an operation on 1024 elements with each operation taking 5 clock cycles. How many cycles are needed to complete the operation?

- (A) 64 (B) 80
(C) 100 (D) 128

Q89 Given a project that uses the COCOMO model with an estimated effort of 2000 person-months and a productivity rate of 5 person-month per KLOC, what is the estimated size of the project in KLOC ?

- (A) 200 (B) 400
(C) 100 (D) 50

Q90 Which of the following is context free language?

- (A). $\{w^i x^j y^k z^l \mid i + k = j + l, \text{ where } i, j, k, l \geq 0\}$
(B). $\{w^i x^j y^k z^l \mid i = j, k = l, \text{ where } i, j, k, l \geq 0\}$
(C). $\{w^i x^j y^k z^l \mid i = j = k \text{ and } k \neq l, \text{ where } i, j, k, l \geq 0\}$
(D). $\{w^i x^j y^k z^l \mid i = j = k + l, \text{ where } i, j, k, l \geq 0\}$
(E). $\{w^i x^j y^k z^l \mid i = j = l \text{ and } k \neq l \text{ where } i, j, k, l \geq 0\}$

- (A) (A) and (B) Only
(B) (B) and (C) Only
(C) (C) and (D) Only
(D) (D) and (E) Only

Directions (91-95) Read the following passage and answer the given questions.

Read the below passage and answer the questions .

Artificial Neural Networks (ANNs) are computational models inspired by the human brain's neural networks. They consist of interconnected nodes, or neurons, organized into layers : an input layers, one or more hidden layers and an output layers. Each connection between neurons has a weight that adjusts as learning progresses allowing the network to

adopt and improve its performance. ANNs are particularly effective in recognizing patterns making them valuable for tasks such as image and speech recognition, Natural language processing and predictive analytics. Learning in ANNs typically involves training algorithms like back propagation, which minimize the error by adjusting the weights. As a subset of machine learning, ANNs have revolutionized the field of Artificial Intelligence by providing solutions to complex problems that traditional algorithms struggle with.

Q91 What is the role of Back Propagation Algorithm?

- (A) To reduce error
(B) To secure network
(C) To control speed of data
(D) To add different layers

Q92 What is the role of weights in an ANN ?

- (A) To store data
(B) To adjust and improve network performance
(C) To control the speed
(D) To secure the network

Q93 Which of the following is/are the application area(s) of ANN ?

- (A) Natural Language Processing
(B) Image Processing
(C) Pattern Recognition
(D) Speech Recognition

Choose the correct answer from the options given below :

- (A) (A) and (B) Only
(B) (B) and (C) Only
(C) (A), (B) and (C) Only
(D) (A), (B), (C) and (D)

Q94 Artificial Neural Networks (ANNs) are inspired by :

- (A) Quantum mechanics
(B) Human brain's neural network
(C) Computer Hardware architecture
(D) Genetic algorithm



Q95 Which of the following layers may be more than one in number ?

- (A) Input layer (B) Hidden layer
(C) Output layer (D) Physical layer

Directions (96-100) Read the following passage and answer the given questions.

Read the below passage and answer the questions .

The Banker's Algorithm is a critical deadlock avoidance method in operating systems, designed to facilitate resource allocation without causing deadlock. It operates by maintaining information about the maximum resources. Each process may require, the current allocated resources and the available resources in the system. The algorithm checks each resource request to determine, if granting it would leave the system in a safe state, meaning that there is always a sequence in which all processes can complete their execution without getting stuck due to resource unavailability. Each process must specify its maximum demand for each resource type before it starts execution. When a process requests additional resources, the algorithm checks if granting the request will keep the system in a safe state. If so, the resources are allocated otherwise the process must wait until its request can be safely fulfilled.

Q96 Which of the following is NOT a requirement for Banker's algorithm to grant a resource request ?

- (A) The requested resources must be available.
(B) The system must be in a safe state after granting the request
(C) The request must not exceed the maximum resources the process can request.

(D) The process must be the only one requesting resources.

Q97 Which data structure does the Banker's Algorithm use to maintain the state of available, maximum and allocated resources ?

- (A) Priority Queue
(B) Hash table
(C) Wait-for-Graph
(D) Matrices and Vectors

Q98 What is the significance of the Banker's algorithm in terms of resource management?

- (A) It ensures that all processes can finish their execution without deadlock.
(B) It eliminates the need for processes to request resources.
(C) It accelerates the execution of critical sections in processes.
(D) It minimizes the number of context switches between processes.

Q99 What is the primary goal of the Banker's Algorithm?

- (A) To allocate resources optimally
(B) To prevent processes from requesting resources
(C) To detect and recover from deadlocks
(D) To maximise CPU utilization

Q100 What information is used to determine if a resource request can be granted ?

- (A) Available resources and current allocation of each process
(B) CPU utilization of each process
(C) Number of processes waiting for resources
(D) Arrival time of each process



Answer Key

Q1 (C)
Q2 (A)
Q3 (B)
Q4 (A)
Q5 (A)
Q6 (D)
Q7 (D)
Q8 (D)
Q9 (B)
Q10 (B)
Q11 (B)
Q12 (D)
Q13 (C)
Q14 (B)
Q15 (A)
Q16 (A)
Q17 (D)
Q18 (C)
Q19 (B)
Q20 (D)
Q21 (A)
Q22 (C)
Q23 (D)
Q24 (C)
Q25 (C)
Q26 (C)
Q27 (A)
Q28 (B)
Q29 (A)
Q30 (B)
Q31 (B)

Q32 (D)
Q33 (D)
Q34 (C)
Q35 (A)
Q36 (A)
Q37 (B)
Q38 (C)
Q39 (A)
Q40 (B)
Q41 (A)
Q42 (A)
Q43 (C)
Q44 (A)
Q45 (B)
Q46 (A)
Q47 (A)
Q48 (A)
Q49 (C)
Q50 (B)
Q51 (A)
Q52 (C)
Q53 (C)
Q54 (C)
Q55 (C)
Q56 (D)
Q57 (B)
Q58 (C)
Q59 (D)
Q60 (C)
Q61 (A)
Q62 (B)



Q63 (C)
Q64 (C)
Q65 (D)
Q66 (B)
Q67 (C)
Q68 (C)
Q69 (C)
Q70 (A)
Q71 (B)
Q72 (D)
Q73 (C)
Q74 (B)
Q75 (C)
Q76 (B)
Q77 (A)
Q78 (A)
Q79 (A)
Q80 (B)
Q81 (B)

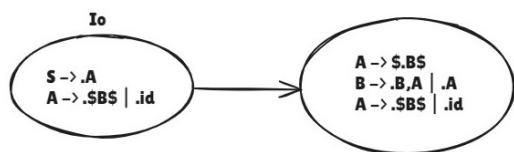
Q82 (A)
Q83 (C)
Q84 (A)
Q85 (B)
Q86 (C)
Q87 (A)
Q88 (A)
Q89 (B)
Q90 (A)
Q91 (A)
Q92 (B)
Q93 (D)
Q94 (B)
Q95 (B)
Q96 (D)
Q97 (D)
Q98 (A)
Q99 (A)
Q100 (A)



Hints & Solutions

Q1 Text Solution:

The correct answer is **Option C (5)**.



GOTO(I₀, \$)

A -> \$.B\$

B -> .B,A

B -> .A

A -> \$.B\$

A -> .id

The number of items in the set GOTO(I₀, \$) is 5.

Q2 Text Solution:

The correct answer is **Option A (In 1NF, but not in 2 NF)**.

The minimum number of attributes that uniquely identify tuples in a table is called the **Candidate Key**. Candidate keys may have multiple attributes.

Normal forms are used to minimize redundancy from a set of relations.

Normal Forms (NF)	Description
1NF	If a relation contains only atomic values then it is in 1NF.
2NF	If a relation is in 1NF and all non-key attributes are fully functionally dependent on the primary key.
3NF	If a relation is in 2NF then there must not exist any transitive dependency (i.e. If $X \rightarrow Y$ and $Y \rightarrow Z$ then $X \rightarrow Z$).
BCNF	The table should be in 3NF and X should be a superkey for every functional dependency $X \rightarrow Y$ in a given relation.

For given schema R(A, B, C, D, E, F, G, H), each field of R contains only atomic values (i.e. 1NF).
 $F = \{CH \rightarrow G, A \rightarrow BC, B \rightarrow CFH, E \rightarrow A, F \rightarrow EG\}$ is a set of functional dependencies.

Since D is not a part of any functional dependency so either it can be a candidate key or may be a part of candidate key.

First, find all possible candidate keys using closure.

$(AD)^+ = \{ABCDEFGH\}$

$(BD)^+ = \{ABCDEFGH\}$

$(ED)^+ = \{ABCDEFGH\}$

$(FD)^+ = \{ABCDEFGH\}$

These are the only candidate keys that are possible.

Prime Attributes (i.e. part of key attributes) = A, B, D, E, F

Non-Prime Attributes (i.e. attributes that are not part of the key) = C, G, H

$A \rightarrow BC, B \rightarrow CFH, F \rightarrow EG$, Here we can clearly see that partial dependency exists i.e. a non-prime attribute (i.e., not part of any candidate key) is functionally dependent on only a part of the primary key, rather than the entire primary key. The relation is not in 2NF.

Q3 Text Solution:

The correct answer is **Option B ((A) and (B) Only)**.

A Pointer is a variable whose value is the address of another variable.

(A) Pointers can be used to access array

elements: True. Suppose arr is a 2-D array, we can access any element $arr[i][j]$ of the array using the pointer expression $((arr + i) + j)$.

(B) Pointers can store the address of another

pointer: True. Pointers are variables, but instead of holding the data directly, they hold addresses that point to the location in memory where the data resides.

(C) Pointers are automatically dereferenced in the expression: False. We have to use the



dereference operator (*) to get the value.

(D) Pointers cannot be used to access structure members: False. Structure pointers are used to indirectly access and modify the members of a structure by pointing to the structure's memory address.

Q4 Text Solution:

The correct answer is **Option A (20 m)**.

To determine the 1-bit delay in a token ring network, we need to consider the transmission speed and propagation speed.

Transmission speed (T_s) = 10^7 bits per second (bps)

Transmission delay for 1 bit = $1 / T_s = 1/10^7 = 0.1 \mu s$

Propagation speed (P_s) = 200 meters per microsecond (m/ μs)

1-bit delay = (Transmission delay for 1 bit) x (Propagation speed)

$$\begin{aligned} &= [(1 \text{ bit}) / T_s] \times P_s \\ &= [(1 \text{ bit}) / (10^7 \text{ bps})] \times (200 \text{ m}/\mu s) \\ &= (0.1 \mu s) \times 200 \text{ m}/\mu s \\ &= 20 \text{ m} \end{aligned}$$

Therefore, the 1-bit delay in this network is equivalent to 20 meters.

Q5 Text Solution:

The correct answer is **Option A ((A), (B), (C) Only)**.

Mutual exclusion is a concurrency control technique that prevents multiple threads from accessing the same shared resource at the same time. Critical section is a shared resource, such as a global variable, database table, or network server file, that should not be accessed by more than one thread at a time

(A) True : Mutual exclusion ensures that only one process can be in a critical section at any given time.

(B) True : Mutual exclusion are designed to prevent conflicts and ensure that only one process can access shared resources at a time.

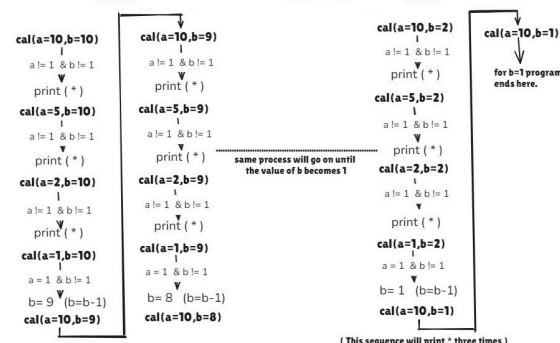
(C) True : Mutual exclusion can use various algorithms to ensure that processes do not enter

the critical section simultaneously.

(D) False : Mutual exclusion allows multiple processes to access the critical section simultaneously to improve performance.

Q6 Text Solution:

The correct answer is **Option D (27)**.



every time when this sequence run it prints the * three times.

This sequence will run 9 times (from b = 10 to b = 2). So

9(Number of time sequence run) x 3(Number of * printed in each sequence) = 27

Q7 Text Solution:

The correct answer is **Option D ((C) and (E) Only)**.

A reflexive relation is a binary relation on a set where every element of the set is related to itself. In a reflexive relation, $(a, a) \in R$ for every element, where a is the element, A is the set, and R is the relation.

(A) $R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$ is not a reflexive relation because $(3,3) \notin R_1$

(B) $R_2 = \{(1,1), (1,2), (2,1)\}$ is not a reflexive relation because Here $(2,2) \notin R_2$

(C) $R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$ is a reflexive relation because every element is related to itself.

(D) $R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$ is not a reflexive relation because $(1,1)$, $(2,2)$, $(3,3)$ and $(4,4) \notin R_4$. No element is related to itself.

(E) $R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$ is a reflexive relation because every element is related to itself.



Q8 Text Solution:

Step 1: Identify which attributes are dependent on others.

$U \rightarrow V$ means if we know U, we can determine V.

$VW \rightarrow X$ means knowing V and W, we can determine X.

$Y \rightarrow W$ means knowing Y, we can determine W.

$X \rightarrow U$ means knowing X, we can determine U.

Step 2: Find the closure of attribute combinations to determine candidate keys.

Closure of UY:

$U \rightarrow V$, so UY determines V.

$Y \rightarrow W$, so UY determines W.

$VW \rightarrow X$, so UY determines X.

$X \rightarrow U$, so UY already determines U (nothing new).

Thus, $UY^+ = \{U, V, W, X, Y\}$, covering all attributes except Z.

Closure of VY:

$Y \rightarrow W$, so VY determines W.

$VW \rightarrow X$, so VY determines X.

$X \rightarrow U$, so VY determines U. Thus, $VY^+ = \{U, V, W, X, Y\}$, covering all attributes except Z.

Closure of XY:

$X \rightarrow U$, so XY determines U.

$U \rightarrow V$, so XY determines V.

$Y \rightarrow W$, so XY determines W. Thus, $XY^+ = \{U, V, W, X, Y\}$, covering all attributes except Z.

Step 3: Evaluate the candidate key options.

To cover all attributes, including Z, we need to extend the combinations:

UY, VY, and XY determine all attributes except Z.

Therefore, adding Z to these combinations will cover all attributes.

Thus, the correct candidate keys are UYZ, VYZ, and XYZ.

The correct option is D: UYZ, VYZ, XYZ.

Q9 Text Solution:

The correct way to declare a function pointer in C is:

Option B: `int (*func)(int, int);`

Explanation:

In this declaration, `(*func)` indicates that func is a pointer to a function.

The function takes two int arguments (int, int)

and returns an int.

Let's break down the other options:

A: `int *func(int, int);`

This declares a function named func that takes two int arguments and returns a pointer to an int.

This is not a function pointer declaration.

C: `int (func*) (int, int);`

This syntax is incorrect because the asterisk * should be inside parentheses to indicate a pointer.

D: `int *func* (int, int);`

This is also invalid due to incorrect placement of asterisks. The syntax doesn't follow the proper rules for declaring a function pointer.

Q10 Text Solution:

The correct sequence of steps in the Feature Driven Development (FDD) process is:

(A) Develop an overall model $\rightarrow$ (E) Build a feature list $\rightarrow$ (C) Plan by feature $\rightarrow$ (D) Design by feature $\rightarrow$ (B) Build by feature.

Explanation:

Feature Driven Development (FDD) is an iterative and incremental software development methodology with the following sequence of steps:

Develop an overall model: Understand the domain and develop a high-level model.

Build a feature list: Identify and organize all the necessary features.

Plan by feature: Create a development plan based on the feature list.

Design by feature: Design individual features before implementation.

Build by feature: Implement the features in small iterations.

Q11 Text Solution:

The number 45_y is given in an unknown base y, and we need to determine its possible values.

Interpretation:

The number 45_y means $4 \times y + 5$ in base y, where both digits (4 and 5) must be valid digits in base y.

In base y, digits range from 0 to $y-1$. Therefore,



for the number 45y to be valid, y must be greater than 5 (since 5 is the highest digit used in the number).

Evaluation of each base:

Base 5: Invalid, because the digit 5 is not allowed in base 5 (the maximum digit is 4).

Base 6: Valid, since digits 4 and 5 are allowed in base 6 (digits 0-5 are valid).

Base 7: Valid, since digits 4 and 5 are allowed in base 7 (digits 0-6 are valid).

Base 8: Valid, since digits 4 and 5 are allowed in base 8 (digits 0-7 are valid).

Conclusion:

The possible values for y are 6, 7, and 8.

Thus, the correct answer is B: (B), (C), and (D) only.

Q12 Text Solution:

Breakdown of the array x[]:

$$x[0] = 2$$

$$x[1] = 4$$

$$x[2] = 6$$

$$x[3] = 8$$

$$x[4] = 10$$

Pointer y:

y points to x[4], which is 10.

Iteration by Iteration:

First Iteration (a = 0):

$$*y = 10$$

$$*(y - a) = *(y - 0) = *y = 10$$

$$(*y - a) = (10 - 0) = 10$$

$$*(y - a) = 10$$

$$\text{Result: } b = b + (10 - 10) = b + 0 = 0$$

Second Iteration (a = 1):

$$*y = 10$$

$$*(y - a) = *(y - 1) = x[3] = 8$$

$$(*y - a) = (10 - 1) = 9$$

$$*(y - a) = 8$$

$$\text{Result: } b = b + (9 - 8) = b + 1 = 1$$

Third Iteration (a = 2):

$$*y = 10$$

$$*(y - a) = *(y - 2) = x[2] = 6$$

$$(*y - a) = (10 - 2) = 8$$

$$*(y - a) = 6$$

$$\text{Result: } b = b + (8 - 6) = b + 2 = 3$$

Fourth Iteration (a = 3):

$$*y = 10$$

$$*(y - a) = *(y - 3) = x[1] = 4$$

$$(*y - a) = (10 - 3) = 7$$

$$*(y - a) = 4$$

$$\text{Result: } b = b + (7 - 4) = b + 3 = 6$$

Fifth Iteration (a = 4):

$$*y = 10$$

$$*(y - a) = *(y - 4) = x[0] = 2$$

$$(*y - a) = (10 - 4) = 6$$

$$*(y - a) = 2$$

$$\text{Result: } b = b + (6 - 2) = b + 4 = 10$$

Final Output:

Option D: 10

Q13 Text Solution:

List I - List II:

(A) The activation function $\rightarrow$ (II) is one of the key components of the perceptron as in the most common neural network architecture.

The output of the perceptron plays a crucial role in determining the decision boundaries of neural networks

(B) The learning method of perceptron $\rightarrow$ (I) is called the delta rule.

The delta rule (also known as the perceptron learning rule) is used to adjust weights in neural networks.

(C) Areas of application of artificial neural $\rightarrow$ (IV) system identification and control.

Neural networks are widely used in system identification, control systems, and various other applications.

(D) The output of the perceptron $\rightarrow$ (III) is always boolean like a switch

This is true for the activation function in a perceptron, which uses a threshold function for binary classification.

Correct Match:

(A) $\rightarrow$ (II)

(B) $\rightarrow$ (I)

(C) $\rightarrow$ (IV)

(D) $\rightarrow$ (III)

Thus, the correct answer is Option C.



Q14 Text Solution:

Correct Matching:

(A) Representation of bits matches with (II)

Physical layer: The physical layer is responsible for the transmission and reception of raw bitstreams over a physical medium.

(B) Physical Address matches with (IV) Data

link layer: The data link layer is responsible for addressing and access control on the local network, typically using MAC addresses (physical addresses).

(C) Logical Address matches with (III) Network

layer: The network layer handles logical addressing, such as IP addresses, used for routing data between different networks.

(D) Segmentation and reassembly matches with

(I) Transport layer: The transport layer (e.g., TCP) handles the segmentation of data into packets for transmission and reassembles them at the destination.

Correct Answer:

Option B: (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

Q15 Text Solution:

The correct answer is Option A: (A) and (C) Only.

Extreme Programming (XP):

XP is an Agile process model that is focused on improving software quality and responsiveness to changing customer requirements.

Key practices include pair programming, continuous integration, and test-driven development.

Scrum:

Scrum is another Agile process model that divides work into small, manageable chunks called "sprints," typically lasting 2-4 weeks. It focuses on collaboration, accountability, and iterative progress toward a well-defined goal.

Waterfall:

Waterfall is a traditional, linear project management model, where each phase (requirement gathering, design, implementation, testing, and maintenance) happens sequentially. It is not Agile, as it lacks the flexibility to adapt to

changes once the project progresses to later stages.

Spiral:

The Spiral model combines iterative development with systematic aspects of the Waterfall model.

It focuses on risk analysis and continuous refinement but isn't inherently Agile, although it has some iterative elements.

Incremental:

The Incremental model breaks the project into small pieces that are developed and delivered incrementally.

While it uses some iterative aspects, it is not strictly considered an Agile process.

Q16 Text Solution:

Statement (A): It can only be executed by the Operating System kernel and not by user applications.

True: Privileged commands can only be used in kernel mode (supervisor mode). User programs in user mode can't use these commands directly because they might interfere with important system functions.

Statement (B): It is designed to perform operations directly affecting the hardware or system state such as I/O operations or changing memory management settings.

True: Privileged instructions handle tasks like managing I/O and memory, which need to be done by the operating system's kernel to keep the system stable.

Statement (C): User applications can execute privileged instructions if they have correct permissions, set by the Operating System.

False: User programs can't run privileged instructions directly.

Statement (D): It is usually executed in user mode to ensure the safety and security of the system.

False: User mode can't access privileged instructions. These instructions run in kernel mode to ensure the system remains secure.



Allowing user programs to access them could compromise system integrity.

The answer is option A where Statements A and B are True.

Q17 Text Solution:

In a CSMA/CD (Carrier Sense Multiple Access with Collision Detection) network, the minimum frame size is determined to detect a collision before the frame is completely transmitted.

Data rate: 100 Mbps (Megabits per second)

Minimum frame size: 1250 bytes = $1250 \times 8 = 10,000$ bits

Distance: 1 km

Now, the time to transmit the minimum frame is:

Transmission time = Frame size (in bits) / Data rate (in bits/sec)

$$= 10,000 \text{ bits} / 100 \times 10^6 = 0.0001 \text{ seconds}$$

For collision detection, the signal must propagate 2 kilometers round-trip from one end of the cable to the other and back (the distance is 2 kilometers).

Thus, the signal speed v in the cable can be calculated as:

$v = \text{Round-trip distance} / \text{Transmission time}$
 $= 2 \text{ km} / 0.0001 \text{ seconds} = 20,000 \text{ km/sec}$

Thus, the signal speed is **20,000 km/sec**, and the correct option is: **D) 20000**

Q18 Text Solution:

To find the following set of R, let's use the following steps:

1. Follow of R from grammar rules:

By definition, the terminals that can come right after a non-terminal in some kind of sentence form are in the following set of the non-terminal.

2. Analyze productions involving R:

In production $Q \rightarrow RS$, S comes immediately after R . Hence, whatever is in the first set of S is added to the follow of R .

The production $S \rightarrow z \mid \epsilon$ means S can either produce z or ϵ . So the first set of S is $\{z, \epsilon\}$.

3. Add to FOLLOW set:

Since S can produce ϵ , we need to look at what

comes after S in the grammar so that it can be added to the following set.

In $T \rightarrow Qx$, x follows Q . So, after Q derives RS , if S goes to ϵ , the terminal x will follow R .

Thus, the following set of R is $\{z, x\}$.

Answer:

The correct option is (C): $\{z, x\}$

Q19 Text Solution:

As an IP datagram travels from host A to host B in a TCP/IPv4 network, some fields in the IP header may change. Let's look at each field:

(A) TTL (Time-to-Live):

True: Each time the datagram passes through a router, the TTL value decreases by one. This means the TTL will be different by the time it reaches its destination.

(B) Checksum:

True: The Checksum must be updated every time the TTL changes because the TTL is part of the header. So, when the datagram arrives at host B, the checksum will be different.

(C) Fragment Offset:

True: The Fragment Offset changes only if the original datagram is fragmented during transmission. If fragmentation occurs, this value will differ from the original datagram.

(D) Source IP:

False: The Source IP (server A's address) stays the same throughout the journey. Routers do not alter the source IP address.

(E) Destination IP:

False: The Destination IP (host B's address) remains unchanged while the datagram is sent. Routers use this address to route the datagram, but they do not modify it.

The correct option is (B): (A), (B) and (C) Only

Q20 Text Solution:

The correct answer is D: (A) and (C) Only.

Explanation:

(A) All production rules are of the form $A \rightarrow xB$ or $A \rightarrow x$, where A and B are non-terminal symbols, and x is a terminal symbol.

This statement is TRUE. A regular grammar has



production rules that are either:

$A \rightarrow xB$ (where A and B are non-terminals, and x is a terminal), or

$A \rightarrow x$ (where x is a terminal).

These rules describe the form of a regular grammar.

(B) Regular grammars are more powerful than context-free grammars and can express any type of language.

False: Regular grammar can only describe regular languages.

Context-free grammars, on the other hand, can represent a wider range of languages, even ones that regular grammars can't.

(C) There is a direct correspondence between regular grammar and finite automata.

True: A comparable finite automaton can recognize a given language for every regular grammar, and vice versa.

(D) Regular grammars can generate languages that are not recognized by any type of automata.

False: Finite automata can identify regular languages produced by regular grammars.

Q21 Text Solution:

We are given the schema $R(P, Q, R, S)$ and the following functional dependencies (FDs):

$P \rightarrow Q$

$Q \rightarrow R$

$R \rightarrow S$

$S \rightarrow Q$

We are decomposing the relation R into the following sub-relations:

$R_1(P, Q)$

$R_2(Q, R)$

$R_3(Q, S)$

We need to check two things:

Preserving all functional dependencies means that, from the decomposed relations, we should be able to enforce the original functional dependencies.

Lossless Join: The decomposition is let us join the subrelations without losing any information.

Step 1: Lossless Join

To ensure that the decomposition allows us to combine the subrelations without losing information, we check for a lossless join.

How to Check:

A lossless join means that when you join the subrelations back together, you get the original relation without extra tuples.

Since the decomposition is based on the functional dependencies and shared attributes (like Q), we can conclude that we can join $R_1 R_2$ and R_3 back together to recreate the original relation R without losing any data.

Step 2: Dependency Preservation

To check if the decomposition is **dependency preserving**, we need to see whether all the original functional dependencies can be enforced from the decomposed relations.

- $P \rightarrow Q$ is preserved in $R_1(P, Q)$
- $Q \rightarrow R$ is preserved in $R_2(Q, R)$
- $R \rightarrow S$ and $S \rightarrow Q$ are indirectly preserved in the combination of R_2 and R_3 because are included in the decomposition.

Since all the functional dependencies can be enforced from the decomposed relations, this decomposition is **dependency-preserving**.

Conclusion

The correct option is (A): Dependency Preserving and lossless join

Q22 Text Solution:

The normal steps in the Agile process are done in small steps and iterations, but there is still a rational order that usually goes like this:

(E) Planning: The team first finds out what they need to work on by defining requirements and breaking jobs down into smaller pieces.

(A) Design: The next step is the design phase, where the architecture and design details are talked over at a high level to make sure everyone on the team knows how to start the development.



(D) Development: Once the team has finished planning and designing, they move on to development, which is where the features are coded and used.

(C) Testing: Once the features have been built, they need to be tested to make sure they work as planned. This is usually done with continuous integration and automatic tests.

(B) Release: At last, the release step consists of delivering the tested and functional elements to the customer or production.

The correct option is C: {(E), (A), (D), (C), (B)}

Q23 Text Solution:

(A) A constructor can be overloaded:

True: In C++, you can have multiple constructors in a class, each with different parameters. This is called constructor overloading, and it allows you to create objects in different ways.

(B) A constructor does not have a return type:

True: Constructors do not have a return type, not even void. Their main purpose is to initialize an object rather than return a value.

(C) A constructor must be declared as a friend function:

False: Constructors do not need to be friend functions. They are usually member functions of the class and serve the purpose of initializing objects.

(D) A constructor is called when an object is destroyed:

False: Constructors are not called when an object is destroyed. They are called when an object is created. Instead, the destructor is the function that runs when an object is destroyed.

The correct option is D: The correct properties are **(A) and (B) Only**.

Q24 Text Solution:

Given:

- p be "It is hot day."
- q be "The temperature is 48°C."

The meaning of $\neg p \wedge \neg q$ in the sentence will be:

- $\neg p$ means "It is not a hot day."
- $\neg q$ means "The temperature is not 48°C."

- $\wedge$ is a logical AND, so $\neg p \wedge \neg q$ means both conditions must be true.

Thus, $\neg p \wedge \neg q$ will be

- "It is neither a hot day nor is the temperature 48°C."

Q25 Text Solution:

Total Lines of Code (LOC) = 2000

Productivity Rate = 10 LOC per person-hour

Person-Hours = Productivity Rate (LOC per person-hour) / Total Lines of Code (LOC)

Substitute the values into the formula:

Person-Hours = 2000/10 = 200 person-hours

The correct option is (C): 200 person-hours are required to complete the project.

Q26 Text Solution:

The steps involved in a Genetic Algorithm (GA) typically follow a structured sequence to evolve solutions to problems. The correct order of the steps is as follows:

(B) Initialisation: To begin, make a start population of possible answers, called individuals.

(E) Evaluation: Use the stated objective function to figure out how fit each person in the population is.

(A) Selection: Choose people based on how competent they are to raise children. More fit people have a better chance of being chosen.

(C) Crossover: To make children, do crossover between certain parents. This is done by combining parts of two-parent answers to make new ones.

(D) Mutation: Change the kids by mutation to make them different in random ways. This keeps the population's DNA diversity high and can help find new solutions in the solution space.

Thus, the correct option is **(C) : (B), (E), (A), (C), (D)**.

Q27 Text Solution:

Explanation:

char *a = "ONE", *b = "TWO";

a points to the string "ONE", and b points to the



string "TWO".

Calling f1(a, b):

Inside f1, the values of x and y are swapped locally using a temporary variable t1. However, since C passes arguments by value, the actual values of a and b in the main() function remain unchanged.

After f1(a, b), a still points to "ONE" and b still points to "TWO".

So the printf statement after f1 prints:

ONE TWO

Calling f2(&a, &b):

Inside f2, the values of *x and *y (i.e., the pointers a and b themselves) are swapped.

After f2(&a, &b), a now points to "TWO" and b points to "ONE".

So the printf statement after f2 prints:

TWO ONE

The correct option is (A): ONE TWO TWO ONE

Q28 Text Solution:

SQL is the standard language for interacting with relational databases, allowing users to perform tasks such as:

Creating databases and tables.

Inserting, updating, and deleting records.

Querying data using SELECT statements.

Defining data structures with commands like CREATE, ALTER, and DROP.

The correct answer is B: To create and manipulate databases.

Q29 Text Solution:

Explanation

Option A: class A { virtual void show() = 0; };

Correct: An abstract class in C++ is defined as a class that has at least one pure virtual function. A pure virtual function is declared by using the syntax virtual function_name() = 0; it indicates that the function must be overridden in derived classes.

Q30 Text Solution:

Time Complexity Analysis:

(A) $T(n) = T(n/2) + 1$

This recurrence relation simplifies to $O(\log n)$.

(B) $T(n) = 2T(n/2) + n$

This results in $T(n) = O(n \log n)$.

(C) $T(n) = 3T(n/3) + n$

$T(n) = O(n \log n)$.

(D) $T(n) = 2T(n/2) + \sqrt{n}$

$T(n) = O(n)$.

(E) $T(n) = T(n-1) + 1$

This simplifies to $T(n) = O(n)$.

Arranging in Increasing Order:

$O(\log n) < O(n) < O(n \log n)$.

Q31 Text Solution:

The correct answer is:

B: By enforcing rules defined in the schema Relational databases ensure data integrity by enforcing constraints, such as primary keys, foreign keys, unique constraints, and not-null constraints, which are defined in the schema. These rules help maintain the accuracy and consistency of data across tables and relationships.

Q32 Text Solution:

The IP address classes based on the first octet (the first number in the IP address) are as follows:

1. **Class A:** IP addresses from **0.0.0.0 to 127.255.255.255**.

Example: **10.20.30.40** → Class A.

2. **Class B:** IP addresses from **128.0.0.0 to 191.255.255.255**.

Example: **180.30.100.10** → Class B.

3. **Class C:** IP addresses from **192.0.0.0 to 223.255.255.255**.

Example: **210.20.30.3** → Class C.

4. **Class E:** IP addresses from **240.0.0.0 to 255.255.255.255** (Reserved for experimental use).

Example: **252.5.15.11** → Class E.

The correct answer is **D: (A)-(III), (B)-(IV), (C)-(II), (D)-(I)**.

Q33 Text Solution:

In descending order of access time (from slowest to fastest), the sequence of storage devices is:



(C) Magnetic Tapes - Slowest access time (seconds or more).
 (E) Optical Disk - Access time in milliseconds, similar to magnetic disks.
 (B) Magnetic Disk - Access time in milliseconds.
 (D) Main Memory - Access time in nanoseconds.
 (A) Registers - Fastest access time (nanoseconds).
 Thus, the correct answer is D: (C), (E), (B), (D), (A).
 NOTE: In UGC NET question paper, the question was asked about ascending order which was wrong. We have changed it to arrange it in descending order.

Q34 Text Solution:

The correct answer is C: Half of the number of vertices.

This result is based on **Dirac's Theorem**, which states that a graph G with n vertices (where $n \geq 3$) is Hamiltonian if the degree of each vertex is at least $\frac{n}{2}$.

Explanation:

- A Hamiltonian graph is a graph that contains a Hamiltonian cycle, a cycle that visits each vertex exactly once.
- Dirac's Theorem provides a sufficient condition for a graph to be Hamiltonian. It states that if every vertex in a graph has a degree $\geq \frac{n}{2}$, where n is the number of vertices, then the graph is Hamiltonian.

Thus, the degree of each vertex should be greater than or equal to half of the number of vertices for the graph to be Hamiltonian.

Q35 Text Solution:

Explanation:

- The Fifth Normal Form (5NF), or Project-Join Normal Form (PJNF), is concerned with Join Dependency.
- A relation is in 5NF if it cannot be decomposed into smaller relations without losing information and if it has no join dependency.

The correct answer is **A: Join Dependency**.

Q36 Text Solution:

The correct answer is **option A**.

- Using the precedence graph method, we can find out Schedule is conflict serializable or not.
- If the precedence graph has any loop then that schedule will not be conflict-serializable.
- If the precedence graph has no loop, then that schedule will be conflict-serializable.

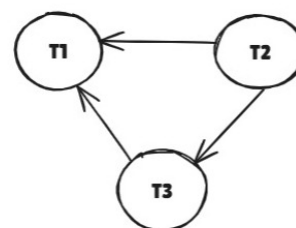
For both schedules, S_1 and S_2 , construct the precedence graphs by identifying conflicts (read-write, write-read, write-write) between transactions.

- For schedule S_1 :

T1	T2	T3
r(x)		
		r(y)
		r(x)
	r(y)	
	r(z)	
		w(y)
		w(z)
r(z)		
w(x)		
w(z)		

•

Precedence Graph



The precedence graph of S_1 has no loop. It is conflict serializable.

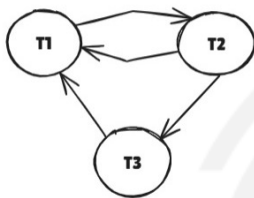
- For schedule S_2 :

T1	T2	T3
r(x)		



		$r(y)$
	$r(y)$	
		$r(x)$
$r(z)$		
	$r(z)$	
		$w(y)$
$w(x)$		
	$w(z)$	
$w(z)$		

Precedence Graph



-
- The Precedence graph of S_2 has loop. It is not conflict-serializable.

Q37 Text Solution:

In a pipelined CPU with 5 stages (Fetch, Decode, Execute, Memory, Write-back), each stage processes one instruction per clock cycle, but instructions can overlap in the pipeline.

The time to complete n instructions in a pipelined processor is given by:

$$\text{total clock cycles} = \text{number of stages} + n - 1$$

Here:

- Number of stages = 5 (F, D, E, M, W)
- Number of instructions $n=10$

So, the total clock cycles required are:

$$5 + 10 - 1 = 14$$

Thus, 14 clock cycles are needed to complete the execution of 10 instructions.

correct option is B).14

**Q38 Text Solution:**

Explanation:

- (A) Half Adder – (IV) 2-bit addition circuit: A half adder is a digital circuit that performs the addition of two binary digits and produces a sum and carry output.
- (B) Decoder – (I) Has n input and 2^n output: A decoder is a combinational circuit that converts n inputs into a unique combination of 2^n outputs, where each output corresponds to one input combination.
- (C) Register – (II) CPU Storage Unit: A register is a small, fast storage location in the CPU used to hold data temporarily during processing.
- (D) Main memory – (III) Used to store programs at runtime: Main memory (RAM) is used to store programs and data while they are being processed by the CPU.

The correct answer is C: (A)-(IV), (B)-(I), (C)-(II), (D)-(III).

Q39 Text Solution:

The correct answer is:

Option A: (A), (B), (D), (C), (E)

1. **(A) Weight Initialization:** Initialize the weights randomly or using a specific method.
2. **(B) Feedforward:** Input data is passed through the network layer by layer to compute predictions.
3. **(D) Loss Calculation:** The loss function calculates the difference between predicted and actual values.
4. **(C) Back Propagation:** The network computes the gradients of the loss function concerning the weights using the chain rule.
5. **(E) Weight Update:** Weights are updated based on the gradients to minimize the loss (usually using gradient descent or its variants).

Q40 Text Solution:

Explanation:

- **(A) Output of combinational circuits depends on only the current input:** This is true because combinational circuits produce



outputs solely based on the current input values without any memory of past inputs.

- **(D) Output of sequential circuits depends on both current input and previous output:** This is true because sequential circuits have memory, and their production depends not only on the current input but also on the previous output (which is stored in memory elements like flip-flops).

The correct answer is:

Option B: (A) and (D) Only

Q41 Text Solution:

Option A

FDs: { $VW \rightarrow X$, $WX \rightarrow V$, $VX \rightarrow W$, $W \rightarrow Y$, $Y \rightarrow Z$ }

This is a **classic case** where BCNF decomposition breaks **dependency preservation**.

Let's find the **candidate key**:

- Try closure of VW :
 - $VW \rightarrow X$
 - $WX \rightarrow V$ (WX is $V + W + X \rightarrow V$, V already there)
 - $VX \rightarrow W$
 - $W \rightarrow Y$
 - $Y \rightarrow Z$
- $\Rightarrow VW^+ = V, W, X, Y, Z \Rightarrow$ So VW is a candidate key

Now check if all FDs are **BCNF compliant** (LHS must be a superkey):

- $VW \rightarrow X$ ✓ (superkey)
- $WX \rightarrow V$ ✗ (WX is not a superkey)
- $VX \rightarrow W$ ✗
- $W \rightarrow Y$ ✗
- $Y \rightarrow Z$ ✗

So BCNF **violated** in multiple FDs.

Try decomposing based on $W \rightarrow Y$ and $Y \rightarrow Z$:

- In the process, **some FDs are lost**, so **dependency preservation fails**.

Conclusion: Cannot decompose into BCNF while preserving dependencies

Correct Answer: A

Q42 Text Solution:

Explanation:

Function Definition: swap

The function `swap(int &x, int &y)` takes two integer references (i.e., addresses) as parameters. This means any changes made to x and y within this function will affect the original variables passed to it. This effectively swaps the values of x and y .

Main Function Execution

1. In the main function, two integers are declared and initialized as $a = 5$ and $b = 10$.
2. In the first call to `swap(a, b)`, where a (which is initially 5) and b (which is initially 10) are passed. After executing this swap, a becomes 10 and b becomes 5.
3. In the second call to `swap(a, b)`, now with updated values ($a = 10, b = 5$), calling swap again will swap them back. After executing this swap, now, a becomes 5 and b becomes 10 again.
4. Finally, after both swaps have been executed, we reach the output statement.

```
cout << a << " " << b;
```

At this point, the value of a is back to 5, and the value of b is back to 10.

Thus, when printed out, it will display the value of a as 5 and the value of b as 10.

The Final Answer is: 5, 10

Q43 Text Solution:

Explanation:

- **(A) Requirement Analysis:** This is a typical activity in the software process lifecycle, where the requirements of the system are gathered and analyzed.
- **(B) System Design:** This is another key activity where the system's architecture and components are designed based on the requirements.
- **(D) Deployment:** After development and testing, deploying the software to a production environment is part of the lifecycle.

The other options are incorrect:



- **(C) Code Refactoring:** While important during software development, it is not a primary activity of the overall software process lifecycle. It is more of a development practice.
- **(E) Substructure:** This is not a recognized phase or activity in the software process lifecycle.

Option C: (A), (B), (D) Only

Q44 Text Solution:

Explanation:

A **relational database** organizes data into tables, where each table consists of rows and columns. Each row represents a record (or tuple), and each column represents an attribute (or field) of the data. This tabular format is the core structure of relational databases, and relationships between tables are defined using keys.

Q45 Text Solution:

(B) and (C) Only

Explanation :-

Coin is tossed successively three times, so sample space is { HHH, HHT, HTT, TTT, TTH, THH, HTH, THT }, hence $n(S) = 8$

We are interested for the event of getting exactly one head or two heads, so event space is { HHT, HTT, TTH, THH, HTH, THT }, hence $n(E) = 6$

To find probability of event E,

$$P(E) = \frac{\text{favourable outcome}}{\text{total number of outcome}}$$

$$\Rightarrow P(E) = \frac{6}{8} = \frac{3}{4}$$

Q46 Text Solution:

The correct answer is **Option A (B, D, A, C).**

The **Karnaugh map (K-map)** is a systematic way of simplifying Boolean expressions. With the help of the K-map method, we can find the simplest expression, which is known as the minimum expression.

The proper sequence for simplifying a Boolean function using a Karnaugh map (K-map) is

(B) Draw the K-map for the given Boolean function

(D) Transfer the truth table values to the K-map

(A) Identify and group the largest possible cluster of 1's

(C) Write the simplified Boolean expression from the grouped clusters

Q47 Text Solution:

The correct answer is **Option A ((A), (B) and (D) only).**

A) $\sim(p \wedge q) \equiv \sim p \vee \sim q$ This statement is always true according to De Morgan's law.

(B) $\sim(p \vee q) \equiv \sim p \wedge \sim q$ This statement is also true according to De Morgan's law.

(C) $p \wedge \sim p \equiv T$ This statement is always false according to Negation law.

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

(D) $\sim(p \Rightarrow q) \equiv p \wedge \sim q$ This is also true.

$p \Rightarrow q$ can be written as $\sim p \vee q$.

$$\sim(p \Rightarrow q) = \sim(\sim p \vee q)$$

$$= \sim(\sim p) \wedge \sim q$$

to De Morgan's law)

$$= p \wedge \sim q$$

(according

(E) $p \vee q \equiv \sim p \vee \sim q$ This is false.

p	q	$p \vee q$	$\sim p$	$\sim q$	$\sim p \vee \sim q$
T	T	T	F	F	F
T	F	T	F	T	T
F	T	T	T	F	T
F	F	F	T	T	T

The Final Answer is: A, B, and D only.

Q48 Text Solution:

The correct answer is **Option A (Base Const Derived Const Derived dest Base dest).**

Execution starts from the main function. In main function, first executes

1.) Base *b = new Derived();

- It creates a new object of type Derived on the heap and assigns its address to a pointer b of type Base*.



- This statement calls the constructor of the Derived class, which in turn calls the constructor of the Base class because the Derived class inherits from the Base class.
- So, the output is "Base Const" followed by "Derived Const".

After that, it executes

2.) delete b;

- It deletes the object pointed to by b, which is of type Derived.
- b is a pointer to a Base class and the Base class has a virtual destructor. A virtual destructor ensures that when an object of a derived class is deleted through a base class pointer, the derived class's destructor is called first.
- So, the destructor of the Derived class is called first, which outputs "Derived dest". Then, the destructor of the Base class is called, which outputs "Base dest".

Combining all outputs from each step gives us:

From constructing objects:

"Base Const" "Derived Const"

From destructing objects (in reverse order):

"Derived dest" "Base dest"

Note: The order of destruction is the reverse of the order of construction. The First Derived class destructor is called, and after that, the Base class destructor is called. So that any resources acquired by the derived class are released before the base class's resources are released.

Q49 Text Solution:

The correct answer is **Option C ($L_1 \cap L_2$ is Recursive Enumerable)**.

Context-free Languages are not closed under intersection.

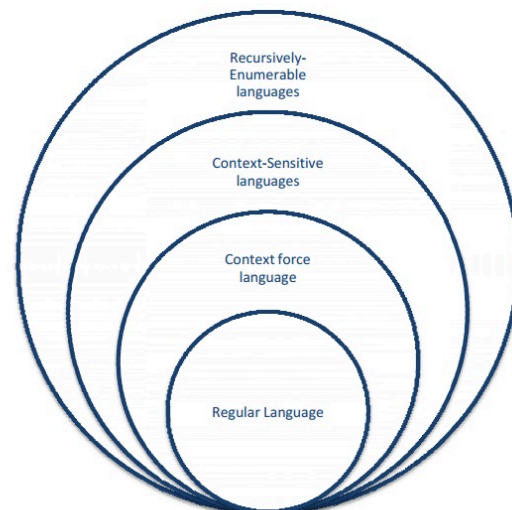


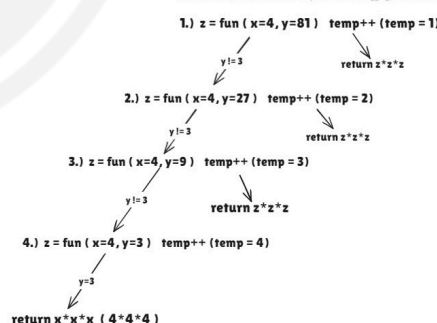
Figure Chomsky Hierarchy of Languages

- $L_1 \cap L_2$ is not context-free. According to the Venn-diagram, if a language is not context-free, it cannot be regular.
- Context-free languages are not closed under intersection. This means that if L_1 and L_2 are CFLs, then $L_1 \cap L_2$ is not necessarily a Context-Sensitive Language (It may or may not be.).

Q50 Text Solution:

The correct answer is **Option B (4)**.

Initially the value of temp = 0.



The final value of temp is 4.

Q51 Text Solution:

The correct answer is **Option A (A, B, C only)**.

A **Syntax tree (also called a parse tree)**

represents a structure of program code in the compiler.

- Syntax trees are useful for a variety of tasks in compiler design, such as type checking, code optimization and code generation. They can



also be used to represent the structure of other types of logical structures, such as natural language sentences or logical expressions.

- Syntax tree can be used for error detection, it determines if a compiler is accurate. If a syntax tree contains an error, the compiler will display an error message. But Syntax tree cannot handle the error it only detects.

Q52 Text Solution:

The correct answer is **Option C** ((A)-(II), (B)-(I), (C)-(IV), (D)-(III)).

List - I	List - II
(A) HTML	(II) Defines the structure of web pages.
(B) DHTML	(I) Allow for dynamic and interactive web pages.
(C) XML	(IV) Used for data storage and transport.
(D) JAVA	(III) Object-oriented programming language for web applications.

(A) HTML stands for Hypertext Markup Language, and it is used to define the structure of the web pages.

(B) DHTML stands for Dynamic Hypertext Markup Language and is a combination of technologies that create interactive and dynamic web pages.

(C) Extensible Markup Language (XML) is a markup language that provides rules to define any data. It is used for data storage and transport.

(D) Java is a Object-oriented programming language and it is used to create a variety of applications and services

Q53 Text Solution:

The correct answer is **Option C** ((A)-(II), (B)-(I), (C)-(IV), (D)-(III)).

List - I (Counter)	List - II (uses/working)
(A) N-bit Ring Counter	(II) Counts exactly N states
(B) Synchronous Counter	(I) Uses universal clock
(C) Asynchronous Counter	(IV) Main clock is applied to first flip-flop only
(D) Decimal Counter	(III) Counts 0 to 9

A Counter is a device which stores or displays the number of times a particular event or process has occurred.

(A) N-bit Ring Counter : The number of states in an N-bit ring counter is N. This is because a ring counter is a shift register with N flip-flops, and the state repeats every N clock cycles.

(B) Synchronous Counter : Synchronous counter has one global clock which drives each flip flop so output changes in parallel.

(C) Asynchronous Counter : In asynchronous counter we don't use universal clock, only first flip flop is driven by main clock and the clock input of rest of the following flip flop is driven by output of previous flip flops.

(D) Decimal Counter : A decimal counter, also known as Modulo-10 counter, has 10 states. It counts from 0 to 9, with each decimal digit represented by a unique binary encoding

Q54 Text Solution:

The correct answer is **Option C** (C, D, B, A).

The correct order of file handling in C is :

- (C) Open the file
- (D) Check for error
- (B) Read from or write to the file
- (A) Close the file

Q55 Text Solution:

The correct answer is **Option C** ((A)-(III), (B)-(IV), (C)-(II), (D)-(I)).



List - I (Algorithms)	List - II (Characteristics)
(A) First Come First Served (FCFS)	(III) Processes are executed in the order they arrive.
(B) Shortest Job First (SJF)	(IV) Select the process for execution with smallest next Burst time.
(C) Round Robin (RR)	(II) Ensures fair allocation of CPU time by assigning time slice.
(D) Priority Scheduling	(I) Each process is assigned a priority.

(A) First Come First Served (FCFS) : FCFS is a scheduling algorithm that processes tasks in the order they are received.

(B) Shortest Job First (SJF) : SJF is a type of scheduling algorithm in which the processor executes the job first that has the smallest execution time.

(C) Round Robin (RR) : Round-Robin scheduling allocates each task an equal share of the CPU time. In its simplest form, tasks are in a circular queue and when a task's allocated CPU time expires, the task is put to the end of the queue and the new task is taken from the front of the queue.

(D) Priority Scheduling : This scheduling algorithm schedules processes according to the priority assigned to each of the processes. Higher priority processes are executed before lower priority processes

Q56 Text Solution:

The correct answer is **Option D ((B), (C), (D), (E), (A))**.

Domain Name Server (DNS) : DNS resolution is the process of resolving a domain name onto an IP address.

following are the steps in a proper sequence for the process of a DNS query :

(B) Query local DNS Server : The browser sends a request to the DNS recursive resolver, which is

generally from the ISP (internet service provider) or a public DNS service.

(C) Local DNS Server Check Cache : Local DNS server check for IP address in cache.

(D) Query root DNS Server : The DNS resolver requests the root DNS server for the IP address.

(E) Query TLD DNS Server : The root server responds with the address of a Top Level Domain (TLD) DNS server.

(A) Query authoritative DNS Server : The resolver requests the TLD DNS server, which responds with the IP address of the domain. The resolver sends a query or request to the DNS, which returns the IP address.

The resolver responds to the browser with the IP address of the requested domain.

Q57 Text Solution:

The correct answer is **Option B ((A)-(III), (B)-(I), (C)-(IV), (D)-(II))**.

List - I (operating system concepts)	List - II. (characteristics)
(A) Paging	(III) Logical to physical mapping
(B) LRU	(I) Evicts least recently used process
(C) C-SCAN	(IV) Circular disk access
(D) Virtual Memory	(II) Extends physical memory

(A) Paging is a storage mechanism, it is used to retrieve processes from secondary memory to main memory as pages.

(B) LRU (Least Recently Used) is an memory management algorithm, it manages the memory by identifying and discarding the least recently used items.

(C) C-SCAN (Circular SCAN) is a disk scheduling algorithm used for disk access.

(D) Virtual Memory enables the execution of programs on machines larger than their physical memory.



Q58 Text Solution:

The correct answer is **option C (2046)**.

Subnet Mask is a 32-bit number which is a sequence of 1's followed by a sequence of 0's where =

NID + SID = Number of 1's (NID =

Network ID and SID = Subnet ID)

HID = Number of 0's (HID = Host ID)

Subnet mask (255.255.248.0) in Binary =

11111111.11111111.11110000.00000000

Number of Hosts per subnet = $2^{(\text{Number of 0's in subnet mask})} - 2 = 2^{11} - 2 = \mathbf{2046}$ (**Hosts per subnet**)

Note :- Every time when we are calculating number of hosts per subnet we subtract 2 because one address is reserved for Network ID and one is reserved for broadcast address.

Q59 Text Solution:

The correct answer is **Option D ((B), (C) and (D) Only)**.

A sorting algorithm is stable if it maintains the relative order of equal elements.

(A) Quick Sort is an unstable sorting algorithm because it does not preserve the relative order of equal elements.

(B) Bubble Sort is a stable algorithm it ensures that two objects with equal keys appear in the sorted output in same order they appear in the input.

(C) Insertion Sort is also a stable algorithm that maintains the relative order of equal elements in the sorted output.

(D) Merge Sort is also a stable algorithm that uses the divide and conquer technique to recursively divide the input into two halves and merge them back together and maintains the relative order of equal elements.

(E) Shell Sort is an unstable sorting algorithm because it does not guarantee that the relative order of equal elements will be maintained in the sorted output.

NOTE: According to the given explanation, **(B), (C) and (D) are correct statements. We have edited the option D because the correct option is not given in the UGC NET exam paper.**

Q60 Text Solution:

The correct answer is Option **C: Red- black tree**

Explanation:

A red-black tree is a self-balancing binary search tree in which the longest path from the root to a leaf is at most twice as long as the shortest path from the root to any other leaf. This balancing promotes efficient operations like insertion, deletion, and lookup.

Q61 Text Solution:

Option : A - $\forall x L(x, Joy)$

Explanation: Given,

Let $L(x, y)$ be the statement "x loves y" where the domain for both x and y consists of all people in the world.

In statement, "Joy is loved by everyone" it implies that **Joy is loved by all the people in the world.**

$\Rightarrow \forall x L(x, Joy)$

In this case, $\forall x$ denotes "for all people x," while $L(x, Joy)$ denotes x's love for Joy.

Q62 Text Solution:

Option: **B (20 50)**

Explanation:

Step 1: First, declare an array.

an array arr of 5 numbers is created by inputting
int arr[5] = {10, 20, 30, 40, 50}. Here, arr[0] = 10

arr[1] = 20 arr[2] = 30 arr[3] = 40

arr[4] = 50.

Step 2: Calculating Pointers

int *p = (int*) (&arr + 1); &arr, which is interpreted as a reference to the array type (i.e., int (*)[5]), provides the address of the entire array arr.

&arr + 1 points to the memory address immediately after the end of the array because adding 1 to &arr causes you to go past the full array.

Step 3: Dereferencing: arr[1], or 20, is obtained by evaluating *(arr + 1).



*(p - 1):P - 1 points to the array's final element, arr[4], because p points to a position that is just past the array's end. 50 is the value of arr[4].

Step 4: Final Result:

Using printf("%d%d", *(arr+1), *(p-1)); the following values will be printed:

20 is the initial value (from *(arr + 1)).

50 is the second value (from *(p - 1)).

Q63 Text Solution:

The correct answer is option **C: (C), (D), (A), (B), (E)**

Explanation:

The sequence of phases in a **Turing Machine (TM)** action during computation is:

(C) Reading a symbol from the tape

(D) Transitioning to a new state based on the current state and symbol read

(A) Writing a symbol on the tape

(B) Moving the tape head left to right

(E) Halting and accepting or rejecting the input

This sequence depicts the logical steps that a Turing Machine takes during the computation process.

Q64 Text Solution:

The correct answer is option C: **(A)-(III), (B)-(I), (C)-(IV), (D)-(II)**.

Explanation:

List - I (software design principles)	List - II (Definition)
(A) Cohesion	(III) Degree to which elements of a module belong together.
(B) Coupling	(I) Degree to which one module relies on another module.
(C) Abstraction	(IV) Simplifying complex reality by modeling classes appropriate to the problem.

(D) Modularity	(II) Dividing a software system into distinct modules.
----------------	--

Cohesion - The degree to which components of a module fit together is known as cohesion.

Coupling - The degree to which one module depends on another is known as coupling.

Abstraction - Simplifying complex reality through problem-appropriate class modeling is known as abstraction.

Modularity- Splitting a software system up into separate modules is known as modularity.

Q65 Text Solution:

Option : D (None of the above)

Explanation:

(A) $L_1 = \{w \in \{0,1\}^* \mid \text{the length of } w \text{ is even}\}$

- **Type:** Regular
- **Recognizability:** Can be recognized by both DFA and NFA.

(B) $L_2 = \{w \in \{0,1\}^* \mid \text{the length of } w \text{ is odd}\}$

- **Type:** Regular
- **Recognizability:** Can be recognized by both DFA and NFA.

(C) $L_3 = \{w \in \{0,1\}^* \mid \text{all 0's come before all 1's in } w\}$

- **Type:** Regular
- **Recognizability:** Can be recognized by both DFA and NFA.

(D) $L_4 = \{w \in \{0,1\}^* \mid w \text{ contains an equal number of 0's and 1's}\}$

- **Type:** Non Regular
- **Recognizability:** neither DFA nor NFA can recognize it because it requires counting.

(E) $L_5 = \{w \in \{0,1\}^* \mid \text{all 1's come before all 0's in } w\}$

- **Type:** Regular
- **Recognizability:** Can be recognized by both DFA and NFA.

NOTE: In the paper of UGC NET None of the options match, So we have edited option D.

Q66 Text Solution:



The correct answer is **B: 37.5**

Explanation: Given a project with an estimated effort of 1500 person-hours and a team of 5 people, each person works 8 hours a day. We have to find out number of days required to complete the project.

Step 1: Determine the team's total person-hours per day:

Total person-hours per day = Number of people × Hours worked per day

Total person-hours per day =

5 people × 8 hours/day = 40 person-hours/day

Step 2: Calculate the total number of days required to complete the project:

Total days = Total estimated effort / Total person-hours per day

Total days = 1500 person-hours / 40 person-hours/day = 37.5 days

Therefore, the correct option is **B**

Q67 Text Solution:

The correct answer is **C: (D), (B), (C), (A)**

Explanation:

The steps of database design should go in the following order:

- **(D) Requirement Analysis**
- **(B) Conceptual Design**
- **(C) Logical Design**
- **(A) Physical Design**

This sequence makes sure that the requirements are understood at the beginning of the database design process, which is then followed by the development of a high-level model, logical refinement, and physical implementation.

Q68 Text Solution:

Option : **C (SYN bit =1, SEQ number =R, ACK bit =1, ACK number =S+1, FIN bit =0)**

Explanation: The following actions are taken during the TCP three-way handshake in order to create a connection:

1. **Sending a TCP segment, host A contains:**

- SEQ number = S
- SYN bit = 1
- Since this is the first connection request, the

ACK bit is set to 0.

- The ACK number is not relevant.

2. **After receiving the SYN segment from host A, host B answers with a segment of its own. In return, Host B is going to:**

- As part of its own SYN segment, set SYN bit to 1 to acknowledge the request.
- Use its own sequence number R.
- To show that it is recognizing A's SYN request, set the ACK bit to 1.
- The ACK number will be set to S + 1

The final TCP segment header information sent by B to A will be: **SYN bit =1, SEQ number =R, ACK bit =1, ACK number =S+1, FIN bit =0**

Q69 Text Solution:

Option: **C (A assumes, B is on same network, but B assumes, A is on a different network)**

At Computer A-A computes its network address using its own IP Address and subnet mask as- 203.197.2.53 AND 255.255.128.0

= 203.197.0.0

A computes the network address of B using IP Address of B and its own subnet mask as- 203.197.75.201 AND 255.255.128.0

= 203.197.0.0

Since both the results are same, so A assumes that B is on the same network.

At Computer B-

B computes its network address using its own IP Address and subnet mask as- 203.197.75.201 AND 255.255.192.0

= 203.197.64.0

B computes the network address of A using IP Address of A and its own subnet mask as- 203.197.2.53 AND 255.255.192.0

= 203.197.0.0

Since both the results are different, so B assumes that A is on a different network.

Thus, Option (C) is correct.

Q70 Text Solution:



The correct answer is option **A: To control the changes made to software**

Explanation:

Software Configuration Management (SCM) uses version control primarily to manage **software changes** and related artifacts over time. This comprises:

1. Monitoring Modifications
2. Cooperation
3. Restored
4. Accountability and Audit
5. Dividing and Combining

Q71 Text Solution:

The correct answer is **B: (A), (C), (B), (D), (E)**

Explanation:

The steps in the compilation process usually go in the following order:

- (A) Lexical Analysis
- (C) Syntax Analysis
- (B) Semantic Analysis
- (D) Intermediate Code Generation
- (E) Code Optimization

This flow illustrates how the compiler breaks down the source code into manageable chunks, examines its organization, confirms its meanings, creates intermediate representations, and then optimizes the code for maximum efficiency.

Q72 Text Solution:

Option - **D ((A)-(IV), (B)-(III), (C)-(I), (D)-(II))**

Explanation :

(A) The number of ways of picking 2 balls out of $6+4=10$ balls is ${}^{10}C_2=45$

The number of ways of getting (2 balls out of 6 white balls) or (2 balls out of 4 red balls) = ${}^6C_2 + {}^4C_2=21$

Required probability = $21/45 = 7/15$

(B) The number of cards in a pack =52 The number of the kings in a pack =4

If we draw one card at random from the 52 cards, Then the probability of getting a king = number

of kings/number of all the cards

i.e. probability of getting a king = $4 / 52$

Or, the probability of getting a king = $1 / 13$

The number of queens in a pack =4

If we draw one card at random from the 52 cards,

Then the probability of getting a queen = number of queens/number of all the cards

i.e. probability of getting a queen = $4 / 52$

Or, the probability of getting a queen = $1 / 13$

The probability of getting a king or a queen out of all the cards = probability of getting a king out of all the cards + probability of getting a queen out of all the cards

= $1 / 13 + 1 / 13$

= $(1+1) / 13$

= $2 / 13$

Hence the probability of getting a king or a queen out of 52 cards is $2 / 13$.

(C) Chance of Drawing One Red Ball and Two White Balls

For an 18-ball bag containing 6 red, 4 white, and 8 blue balls:

The total number of possibilities to select three balls is $({}^{18}C_3)$.

Methods for selecting 1 red = 6C_1

Methods for selecting two white = 4C_2

Total paths = ${}^{18}C_3 = 816$

Ways for 1 red and 2 white = ${}^6C_1 * {}^4C_2 = 6 \times 6 = 36$, **P(C) = $36 / 816 = 3 / 68$**

(D) The likelihood of drawing two blue and one red ball out of a total of eighteen balls

Methods for selecting 2 blue = 8C_2

Methods for selecting 1 red = 6C_1

Methods for 2 blue and 1 red = ${}^8C_2 * {}^6C_1 =$

$28 \times 6 = 168$, **P(D) = $168/816 = 14/68 = 7 / 34$**

Note: We have edited (III) to 2/13

Q73 Text Solution:

The correct option is **C: (A)-(III), (B)-(I), (C)-(IV), (D)-(II)**

Explanation:

Here's the correct matching of List - I with List - II:

List - I	List - II



(A) Natural language processing	(III) A branch of AI focused on understanding and generating human language.
(B) Reinforcement learning	(I) A method of training algorithm by rewarding desired behaviour and/or punishing undesired one.
(C) Support vector machine	(IV) A machine learning technique that finds the hyper plane that best separates different class in a feature space.
(D) Expert system	(II) System designed to emulate the decision making abilities of a human expert.

Q74 Text Solution:**Explanation:**

(A) Dijkstra's Algorithm → (II)

Used to find the shortest path from a single source.

Works only when all edge weights are non-negative.

(B) Floyd–Warshall Algorithm → (I)

Finds shortest paths between all pairs of vertices.

Can handle positive and negative edge weights (no negative cycles).

(C) Bellman–Ford Algorithm → (III)

Finds single-source shortest paths.

Can handle negative edge weights and also detect negative cycles.

(D) Prim's Algorithm → (IV)

Used to find the Minimum Spanning Tree (MST).

Works on connected, weighted, undirected graphs.

The Final Answer is Option B

Q75 Text Solution:**(A), (C) and (D) Only**

The **TRUE** statements about the Process Control Block (PCB) are:

(A) The PCB contains information about the process state, such as whether it is running, waiting or terminated.

(C) The PCB stores the process's memory management information, such as page tables and segment tables.

(D) The PCB is used to track process scheduling information and CPU registers for process execution.

(B) is a **False** statement because the program code and data segments are not part of the PCB.

Q76 Text Solution:

Execution Steps:

Declaration and Initialization:

An array named arr is defined, containing the elements {1, 2, 3, 4, 5}.

A pointer p is initialized to point to the first element of the array (i.e., p points to arr[0], which holds the value 1).

First Print Statement (printf("%d", *p++);):

The expression *p dereferences the pointer p, which at this moment points to arr[0], yielding the value 1.

Following this, the p++ operation shifts the pointer to the next element in the array (now pointing to arr[1], which is 2).

Consequently, the output of this statement is 1.

Second Print Statement (printf("%d", *(p + 1));):

After executing the first print statement, p now references arr[1], which is 2.

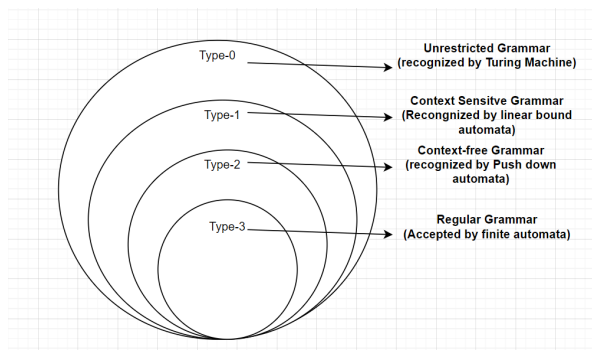
The expression *(p + 1) retrieves the value located one position ahead of the current pointer, leading it to point to arr[2].

The value at arr[2] is 3, so *(p + 1) evaluates to 3. Thus, the output of this statement is 3.

Q77 Text Solution:

(C) < (A) < (B) < (D)





The ascending order of these grammars is
Regular < Context-Free < Context-Sensitive < Unrestricted

The ascending order of the Chomsky Hierarchy is based on the expressive power of the grammars, which refers to the types of languages they can generate or describe. The levels are ordered according to the complexity of the languages they define and the power of the computational models (automata) required to recognize those languages.

Type 3 (Regular Grammars): Can only generate the simplest languages, known as regular languages. These languages have a very restricted structure, such as strings that can be recognized by a finite automaton (like patterns in regular expressions).

Type 2 (Context-Free Grammars): Can generate more complex languages, including many programming languages, like balanced parentheses, that cannot be handled by regular grammars. These are recognized by a pushdown automaton.

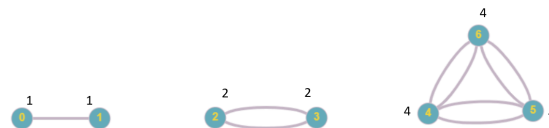
Type 1 (Context-Sensitive Grammars): Can describe even more complex languages, where the structure of the language depends on the context in which the symbols appear. Recognized by a linear bounded automaton.

Type 0 (Unrestricted Grammars): Can describe the most complex and general languages, including all computable languages. These are recognized by a Turing machine, which has no restrictions on memory or structure.

Q78 Text Solution:

Degree of a vertex is given as total no. of edges associated with it. So each edge contributes 1

degree to each vertices which are connected by this edge. For example - 2 vertices connected with 1 edge, degree of graph is 2 ; 2 vertices connected with 2 edge, degree of graph is 4 ; 3 vertices connected with 6 edge, degree of graph is 12.



Therefore, the total sum of degrees of all vertices is twice the number of edges. This can be written as:

$$\text{Sum of degrees of all vertices} = 2 \times \text{Number of edges} = 2l$$

Q79 Text Solution:

Let S be total number of software professionals interviewed i.e. $S = 40$

P be number of people who know Python i.e. $P = 25$

J be number of people who know JAVA i.e. $J = 20$

So $P \cap J$ is the number of people who knew both languages

N be number of people who knew neither languages i.e. $N = 7$

$$P \cup J = S - N = 40 - 7 = 33$$

Applying formula from set theory,

$$P \cup J = P + J - P \cap J$$

$$\Rightarrow 33 = 25 + 20 - P \cap J$$

$$\Rightarrow P \cap J = 45 - 33$$

$$\Rightarrow P \cap J = 12$$



So number of people who know both languages is 12

In ugc net answer key answer is given 20 but the correct should be 12. We have changed the option A.



Q80 Text Solution:**(A), (B), (D) and (E) Only**

(A) In IPv4 addressing, a block of address can be define as $x.y.z.t/n$ in which $x.y.z.t$ define one of the address and $/n$ define the mask. For example - $172.16.45.10/16$ is IPv4 addressing whose subnet mask is $/16$.

(B) The first address in the block can be found by setting the rightmost $32-n$ bits to 0s. For example - $172.16.45.10/16$ is IPv4 addressing whose subnet mask is $/16$. In subnet mask $/16$, there are $32 - 16 = 16$ hosts. Now making 16 right most bits equal to zero we get first network address i.e. $172.16.0.0$

(D) There are five classes in IPv4 address. Five classes are A,B,C,D,E

(E) Supernetting combine several networks into one large network. Supernetting involves aggregating several contiguous IP address ranges into a single IP address range. For example : $192.168.1.0/24$, $192.168.2.0/24$, $192.168.3.0/24$ can be aggregated into a single $192.168.0.0/22$ address range, covering addresses from $192.168.0.0$ to $192.168.3.255$.

(C) is not correct statement. Address of class C is used for Small networks not multicast communication. Multicast communication is done by Class D

Q81 Text Solution:**(A)-(I), (B)-(II), (C)-(IV), (D)-(III)****Correct Table:**

List - I (Testing Type)	List - II (Description)
(A) Unit testing	(I) Testing individual components of the software.
(B) Integrating testing	(II) Testing the interaction between integrated components.

(C) System testing	(IV) Testing the complete system to ensure it meets requirements.
(D) Acceptance testing	(III) Testing to verify the system meets business needs.

Unit Testing: It is the process of checking small pieces of code to ensure that the individual parts of a program work properly on their own. It is used to test individual blocks (units) of functionality. It is done by developers.

Integration Testing: It is performed after unit testing. Once individual components or units have been tested in isolation, they are combined and tested as an integrated group to verify that they work together correctly.

System Testing: This is done to validate the complete and fully integrated software product. It is also known as black-box testing. It is used to evaluate the end-to-end system specification i.e. whether it is meeting all requirements as per SRS or not.

Acceptance Testing: It is the last phase of the testing process. It is done by clients and customers for their satisfaction. The final product is also tested against business needs and regulations of the market.

Q82 Text Solution:**(A) and (C) Only**

(A) Program Counter (PC): This register holds the address of the next instruction to be executed by the processor. It keeps track of the current position in the program.

(B) Instruction Register: This is incorrect because the instruction register holds the instruction currently being executed, not the address of the first instruction.

(C) Accumulator: This register is used to store the result of arithmetic and logic operations performed by the ALU (Arithmetic Logic Unit), so this statement is correct.

(D) Program Counter (PC): This statement is



incorrect because the PC keeps track of the next instruction to be executed, not just the first one.

Q83 Text Solution:

Explanation:

Base Case: If the knapsack's capacity is zero or no items are left, the maximum value is zero. This serves as the initial condition.

Identify Subproblems: Break down the problem by considering each item's inclusion or exclusion. The goal is to find the maximum value for different capacities and item combinations.

Initialize Table: Create a table where rows represent items and columns represent knapsack capacities. This table will store the results of subproblems.

Iterate: For each item and capacity, decide whether to include the item. If included, add the item's value to the best solution for the remaining capacity. Update the table with the maximum value.

Compute Maximum Value: Once the table is filled, the value at the bottom-right cell gives the maximum value for the full capacity of the knapsack.

The Final Answer is: (A), (C), (D), (E), (B)

Q84 Text Solution:

(A) 250

A serial portion of the task cannot be parallelized, so it must be completed by single processes.

Single processes take 200 clock cycles time for the serial portion.

The parallel portion can be parallelized, so it will be divided into 16 processes.

The time required by the single process for the parallel portion is 800 clock cycles.

So time required by 16 processes for the parallel portion:

$$\text{Parallel Time} = 800 \div 16 = 50 \text{ clock cycles}$$

$$\text{Total time} = \text{Serial Time} + \text{Parallel Time} = 200 + 50 = 250 \text{ clock cycles}$$

Q85 Text Solution:

(A), (C), (B), (D)

First IO device(peripheral) sends the DMA request to DMA Controller, then DMAC device sends HOLD signal to CPU. CPU shares starting address from where data transfer should be performed and data count. CPU then sends HLDACK (Hold Acknowledgement) back to DMAC that now information can be passed. Then at last DMAC shares the DMA ACK (DMA Acknowledgement) to the IO device which would successfully let IO device to access or transfer the data.

Q86 Text Solution:

(A)-(I), (B)-(III), (C)-(IV), (D)-(II)

Explanation :-

Master theorem can be used here to determine complexities of recurrence relation. According to this theorem if recurrence relation is of the form :-

$$T(n) = aT(n/b) + \theta(n^k \log^p n)$$

where n = size of the problem

a = number of subproblems in the recursion and $a \geq 1$

n/b = size of each subproblem

$b > 1$, $k \geq 0$ and p is a real number. Then,

1. if $a > b^k$, then $T(n) = \theta(n^{\log_b a})$
2. if $a = b^k$, then
 - (a) if $p > -1$, then $T(n) = \theta(n^{\log_b a} \log^{p+1} n)$
 - (b) if $p = -1$, then $T(n) = \theta(n^{\log_b a} \log \log n)$
 - (c) if $p < -1$, then $T(n) = \theta(n^{\log_b a})$
3. if $a < b^k$, then
 - (a) if $p \geq 0$, then $T(n) = \theta(n^k \log^p n)$
 - (b) if $p < 0$, then $T(n) = \theta(n^k)$

$$(A) T(n) = 2T(n/2) + n$$

Here $a = 2$, $b = 2$ and for $n = n^k \log^p n$,

comparing we get $k = 1$, $p = 0$

$b^k = 2^1 = 2$, so $a = b^k$ and $p > -1$, hence case 2 (a)

$$T(n) = \theta(n^{\log_2 2} \log^{0+1} n) = \theta(n \log(n))$$

$$(B) T(n) = T(n/2) + 1$$

Here $a = 1$, $b = 2$ and for $1 = n^k \log^p n$,

comparing we get $k = 0$, $p = 0$

$b^k = 2^0 = 1$, so $a = b^k$ and $p > -1$, hence case 2(a)

$$T(n) = \theta(n^{\log_1 1} \log^{0+1} n) = \theta(\log(n)), \text{ it}$$

does not match with any.



$$(C) T(n) = 2T(n/2) + 1$$

Here $a = 2$, $b = 2$ and for $1 = n^k \log^p n$,
comparing we get $k = 0$, $p = 0$

$b^k = 2^0 = 1$, so $a > b^k$, hence case 1

$$T(n) = \theta(n^{\log_b a}) = \theta(n^{\log_2(2)}) = O(n)$$

$$(D) T(n) = T(n-1) + n$$

We can solve this question by substitution method,

$$\text{We know } T(n) = T(n-1) + n \text{ ---- (1)}$$

$$\text{so } T(n-1) = T(n-2) + n-1$$

$$T(n-2) = T(n-3) + n-2$$

$$T(n-3) = T(n-4) + n-3$$

$$\begin{array}{r} \cdot \\ \cdot \\ \cdot \end{array}$$

$$T(1) = T(0) + 1$$

We can put substitute values of $T(n-1), T(n-2), \dots, T(1)$ in equation 1

$$T(n) = T(n-1) + n = T(n-2) + 2n-1 = T(n-3) + 3n-3 = \dots$$

After k steps, $T(n)$ becomes:

$$T(n) = T(n-k) + kn - k$$

at base case $n-k = 1 \Rightarrow n=k$

$$T(n) = T(1) + n^2 - n = O(n^2)$$

In ugc net answer key (B) matches with (III) but that is not correct answer.

Q87 Text Solution:

For $\forall x \exists y (x < y)$, option (A) For all real numbers x , there exists a real number y such that x is less than y is correct.

$\forall$ implies "for all"

$\exists$ implies "there exists"

$<$ implies "less than"

Q88 Text Solution:

Option - A (64)

Explanation:

To find out how many cycles a vector processor with 16 lanes will require to finish the operation on 1024 elements:

1. Determine the quantity of operations

required: The number of operations needed is:

Number of operations = 1024 elements / 16 lanes = 64 operations.

2. Find the total number of cycles: The total number of cycles required is:

Total cycles = 64 operations * 5 cycles /

operations = 320 cycles

Number of cycles is 320 and number of operations is 64. In question, number of cycles is asked.

Note: As per UGC, the answer is 64 which is the number of operations.

Q89 Text Solution:

Option - B (400)

Given:

- Approximate work = 2000 person-months - 5 person-months per KLOC is the productivity rate.

2. Find the approximate size in KLOC:

Estimated size (KLOC) = Estimated effort (person-months) / Productivity rate (person-months/KLOC)

Estimated size (KLOC) = 2000

person-months / 5 person-months/ KLOC =

400 KLOC

Q90 Text Solution:

Option - A (A) and (B) Only)

Explanation:

(A) The key constraint $i+k=j+l$ means that the sum of the counts of w and y must equal the sum of the counts of x and z . This kind of balancing can be managed with a stack characteristic of context-free languages.

(B) Two independent conditions on the counts are needed for this, and they can be expressed in a language without regard to context. It is hence context-free

(C) A context-free grammar is unable to capture the dependencies created by the equality condition ($i = j = k$) and the inequality ($k \neq l$). It is therefore not context-free

(D) Context-free grammars are unable to convey the relationship between numerous counts that this language enforces. It is therefore not context-free

(E) As with (C), this sets requirements that, because of the simultaneous equalities and inequalities, result in a language that is not context-free. It is therefore not context-free.

Q91. Text Solution:



Option - **A (to reduce error)**

By computing the gradients of the loss function and adjusting the weights correspondingly during the training phase, the Back Propagation Algorithm's principal purpose is to minimize errors in neural networks.

Q92. Text Solution:

Option - **B (To adjust and improve network performance)**

To find out how much one individual neuron influences another, weights are essential. Through training (e.g., Back Propagation), these weights are modified, allowing the network to learn patterns and enhance prediction performance.

Q93. Text Solution:

Option - **D (A, B, C and D)**

Explanation -

Artificial Neural Networks (ANN) have applications in all of the following fields:

(A) Natural Language Processing: ANNs are frequently employed in applications such as language modeling, sentiment analysis, and machine translation.

(B) Image Processing: Segmentation, object detection, and picture classification are all successfully accomplished by ANNs, particularly Convolutional Neural Networks (CNNs).

(C) Pattern Recognition: ANNs are very good at identifying patterns in data, such as faces or writing digits.

(D) Speech Recognition: ANNs are used in applications such as virtual assistants and transcription services to translate spoken language into text.

In conclusion, all of the options (A, B, C, and D) are accurate and show different ANN application areas.

Q94. Text Solution:

Option - **B (Human brain's neural network)**

ANNs are designed to resemble the architecture and operation of the neural networks seen in the human brain, where neurons interact with one

another (called synapses) and gain knowledge from past experiences.

Q95. Text Solution:

Option - **B (Hidden Layer)**

Explanation -

A. Input layer: The number of neurons in the input layer is usually fixed, matching the number of input features. As a result, while it may include several neurons, it typically only has one layer.

B. Hidden layer: ANNs are capable of having numerous hidden layers, and the depth (number of hidden layers) of the network can change based on the model's complexity (deep learning networks, for example, can have a large number of hidden layers).

C. Output layer: Usually consisting of a single layer, the number of neurons in the output layer might vary. For instance, in a multi-class classification task, there might be one neuron for each class, but the output layer is typically one.

D. Physical layer: It usually refers to layers in network communication protocols.

Q96. Text Solution:

Option - **D (The process must be the only one requesting resources.)**

Explanation:

A: The requested resources must be available:

Before approving a request, the Banker's algorithm verifies that the resources are now available. Should they not be accessible, the request will not be fulfilled.

B: The system must be in a safe state after

granting the request: This is an essential prerequisite. The Banker's algorithm makes sure that the system can continue distribute resources to all processes so that they can all complete, even after approving a resource request. Granting the request could result in a deadlock if the system is not safe.

C: The request must not exceed the maximum

resources the process can request: This is also necessary. To avoid possible resource depletion, the Banker's algorithm verifies that the present request does not surpass the maximum number



of resources that each process is allowed to seek.

D: The process must be the only one requesting resources: This is not necessary. The Banker's algorithm is capable of managing several concurrent resource requests from different programs. It is made to make sure that resource allocation may be controlled in a way that keeps the system safe and prevents deadlock even when there are several processes running at once.

Q97. Text Solution:

Option - **D (Matrices and Vectors)**

Explanation -

A: Priority Queue: This method concentrates on process scheduling according to priority and is not appropriate for managing resource status.

B: Hash table: In the context of the Banker's Algorithm, it does not well illustrate the connections between various operations and the resources they require.

C: Wait-for-Graph: This graph is primarily used to detect deadlocks; it does not keep track of the states of allocated, maximum, and available resources.

D: Matrices and Vectors - Banker's Algorithm makes use of:

- Available Vector: Monitors resources that are accessible.
- Maximum Matrix: Shows the highest level of resources required for every process.
- Allocated Matrix: Displays the resources that are currently assigned to various processes.

Q98. Text Solution:

Option - **A (It ensures that all processes can finish their execution without deadlock.)**

One of the main purposes of the Banker's Algorithm is to guarantee that no process can stall out during its execution. It employs a safety check to make sure that resource distribution does not result in a deadlock, enabling the safe completion of every process.

Q99. Text Solution:

Option- **A (To allocate resources optimally)**

Explanation:

A: To allocate resources optimally - The main objective is to distribute resources so as to prevent deadlocks and guarantee that every process has a chance to finish.

B: To prevent processes from requesting resources - This is untrue; the algorithm permits requests for resources but controls them to avoid deadlocks.

C: To detect and recover from deadlocks - The Banker's Algorithm aids in deadlock prevention, but it does not concentrate on deadlock detection or breakage.

D: To maximize CPU utilization - The Banker's Algorithm prioritizes safe resource management over outright CPU usage optimization, even if effective resource allocation might result in higher CPU utilization.

Q100. Text Solution:

Option - **A (Available resources and current allocation of each process)**

Explanation:

A: Available resources and current allocation of each process - The Banker's Algorithm uses this data to assess whether approving a resource request maintains the system in a safe state.

B: CPU utilization of each process - The Banker's Algorithm bases its resource allocation decisions on demand and availability of resources; CPU usage is not taken into consideration.

C: Number of processes waiting for resources - Although useful for context, this figure by itself does not indicate whether a particular resource request may be approved in a safe manner.

D: Arrival time of each process - The Banker's Algorithm does not take arrival time into account; instead, it concentrates on the allocation and availability of resources at the moment, not the scheduling or timing of processes.

